

XSL simulation using SUN database

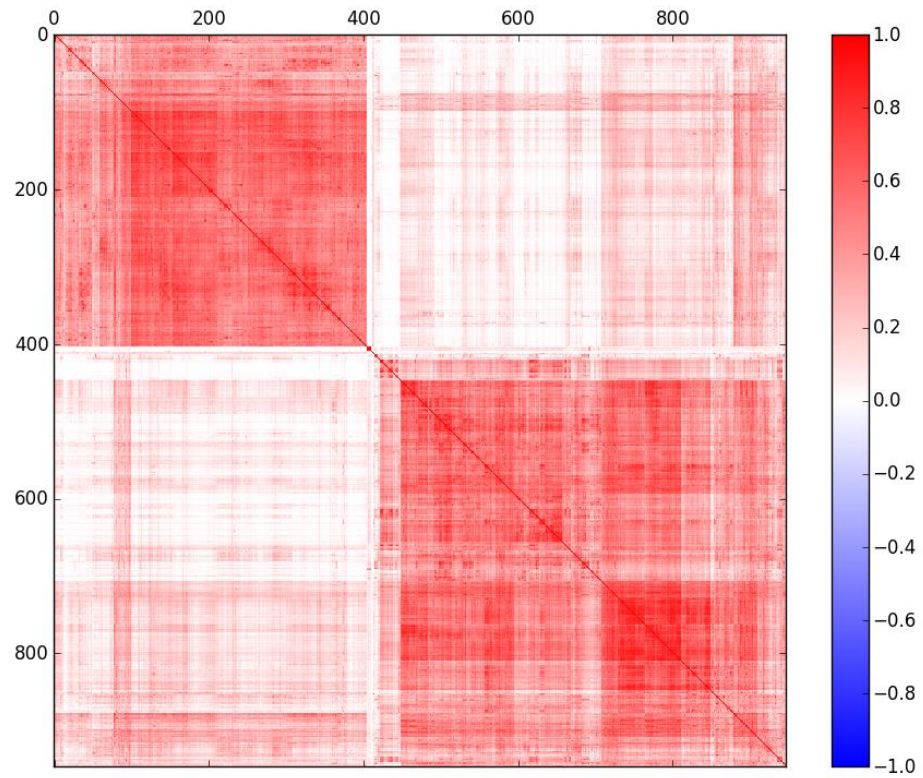
Takuma Torii

20160527

update:20160803

SUN database

SUN database: Full scene correlation



SUN database: Full object correlation



Simulation

Simulation procedure

- Set #words : the max # of words in situation
- Set #objects : the max # of objects in situation
- Until $t < 100,000$
 - Exposure to a situation
 - contains up to #words and #objects
 - these are taken from
 - A scene of SUN database, propto the instance frequency (#images)
 - Uniform randomly; or
 - Zipf randomly (object 1 occurs more)
 - Always include correct word-object pairs
- A right pair (w, o) is learned
 - if $F(w, o) > F(w, a)$
 - where $a \neq o$ is $F(w, a) \geq F(w, b)$ for any $b \neq o$

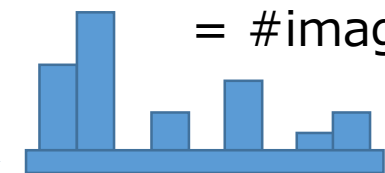
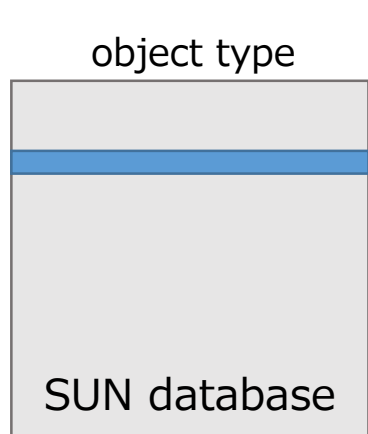
Simulation procedure

Let

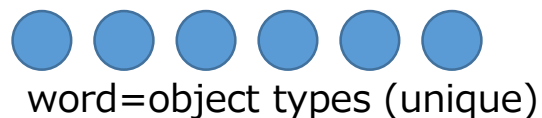
$n = \# \text{words}$

$m = \# \text{objects}$

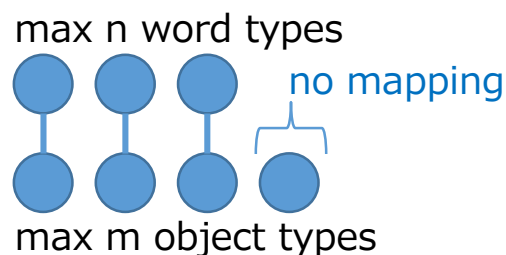
1. choose a scene type and get freq of object types = #images



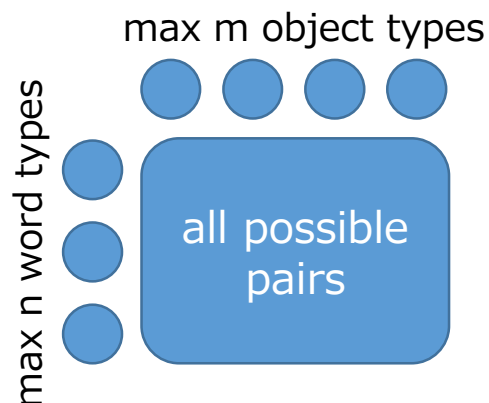
2. sampling $\max(n, m)$ items with replacement $\rightarrow$ uniquify



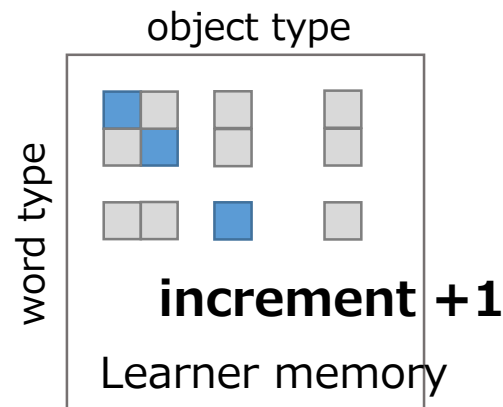
3. take only n and m



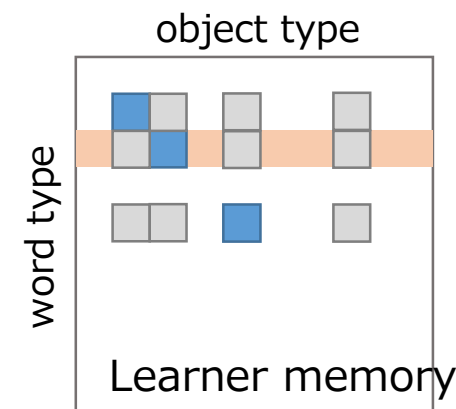
4. learner considers all possible (word, object) pairs



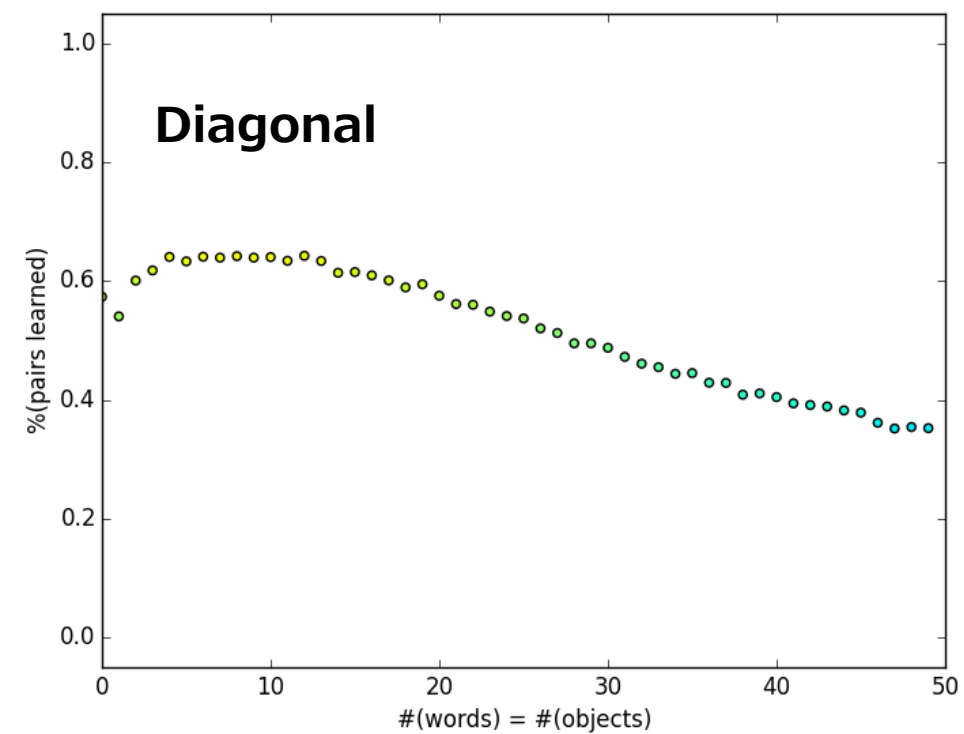
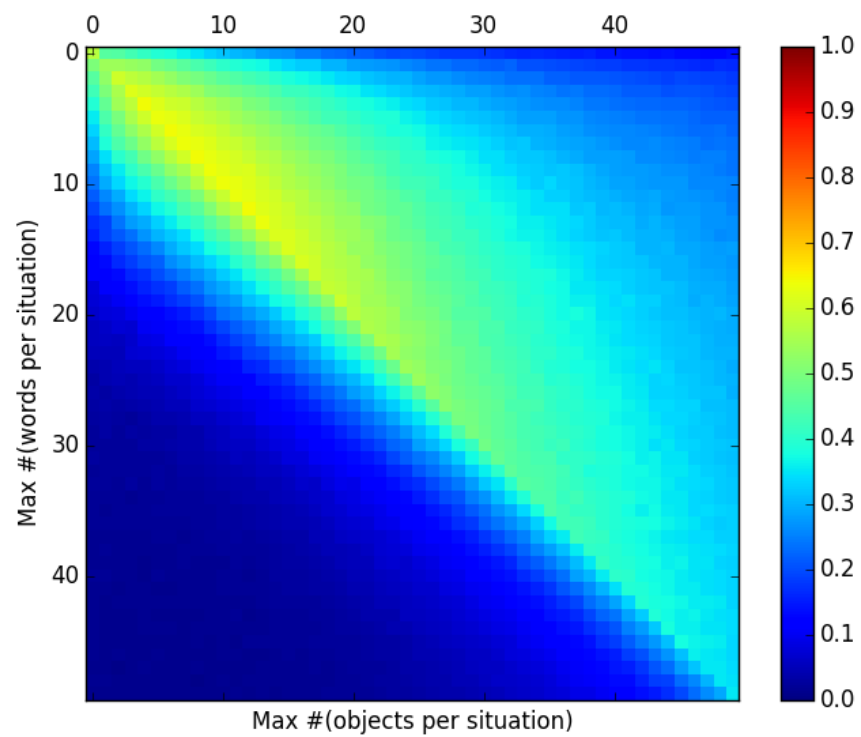
5. increments the freq + 1



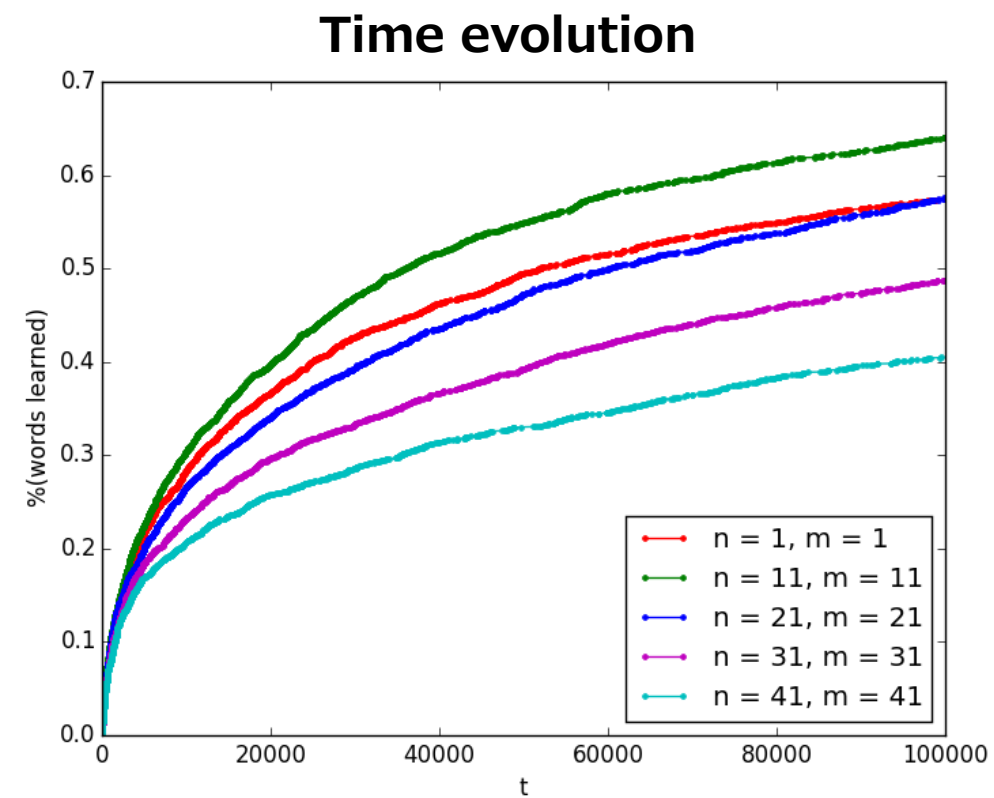
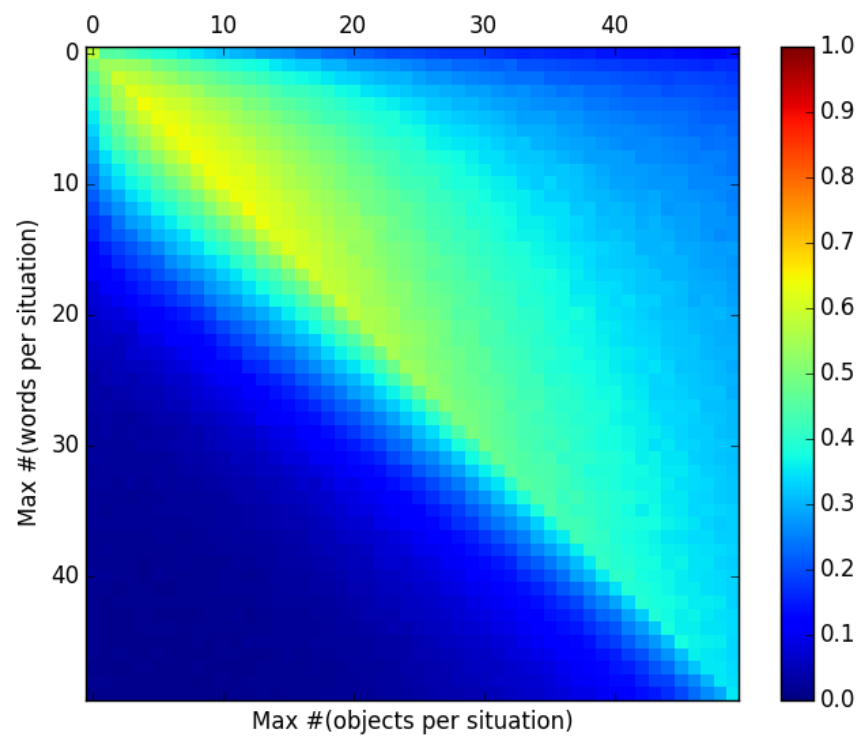
6. check all rows, if $\text{diag} > \max(\text{offdiag})$ then learned



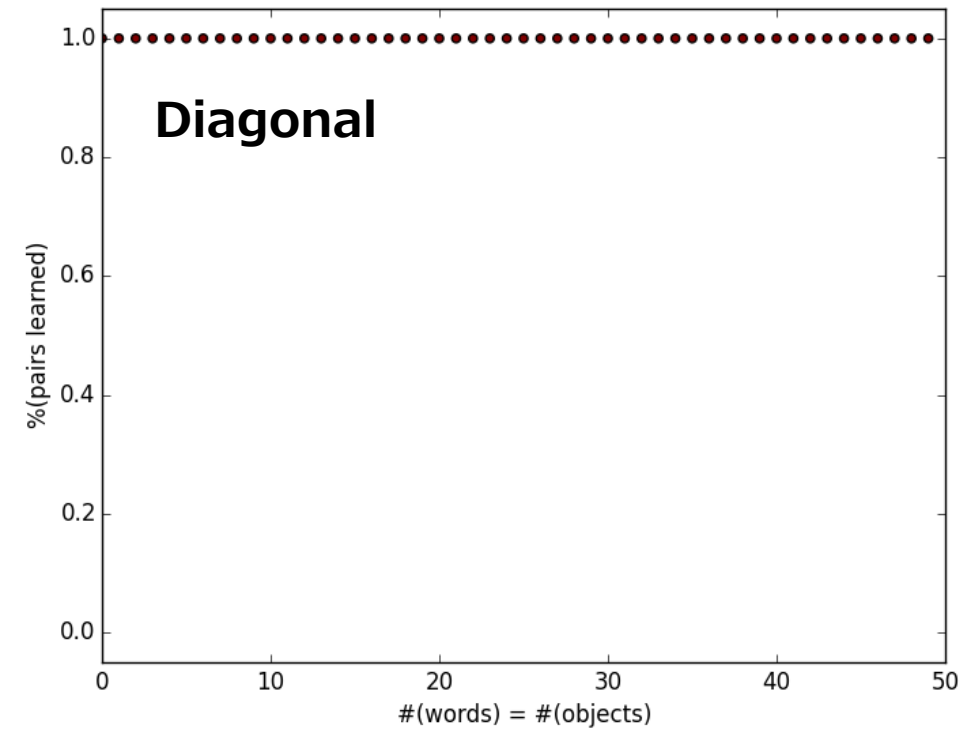
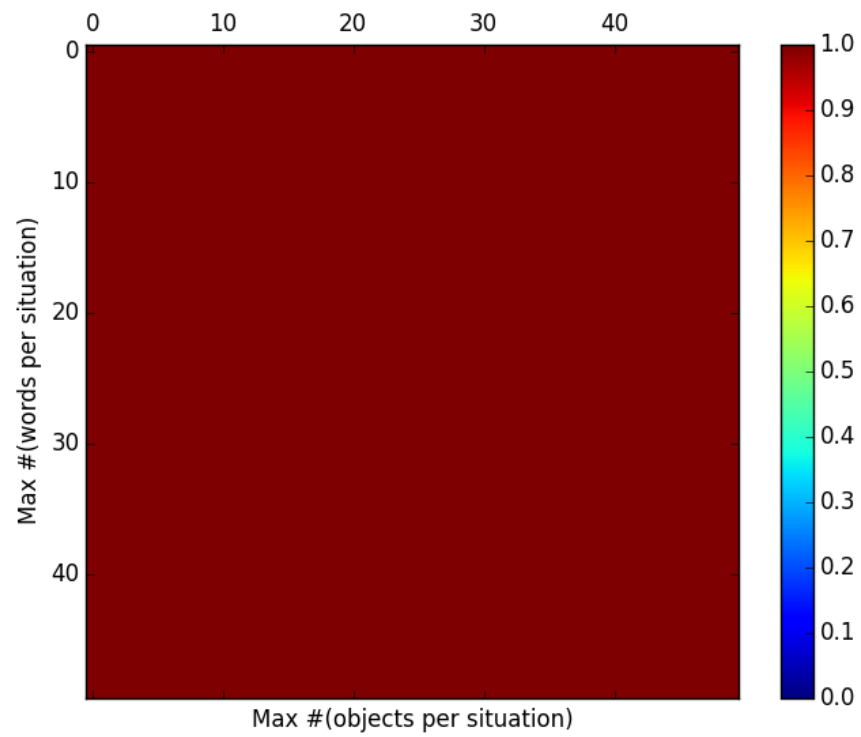
SUN



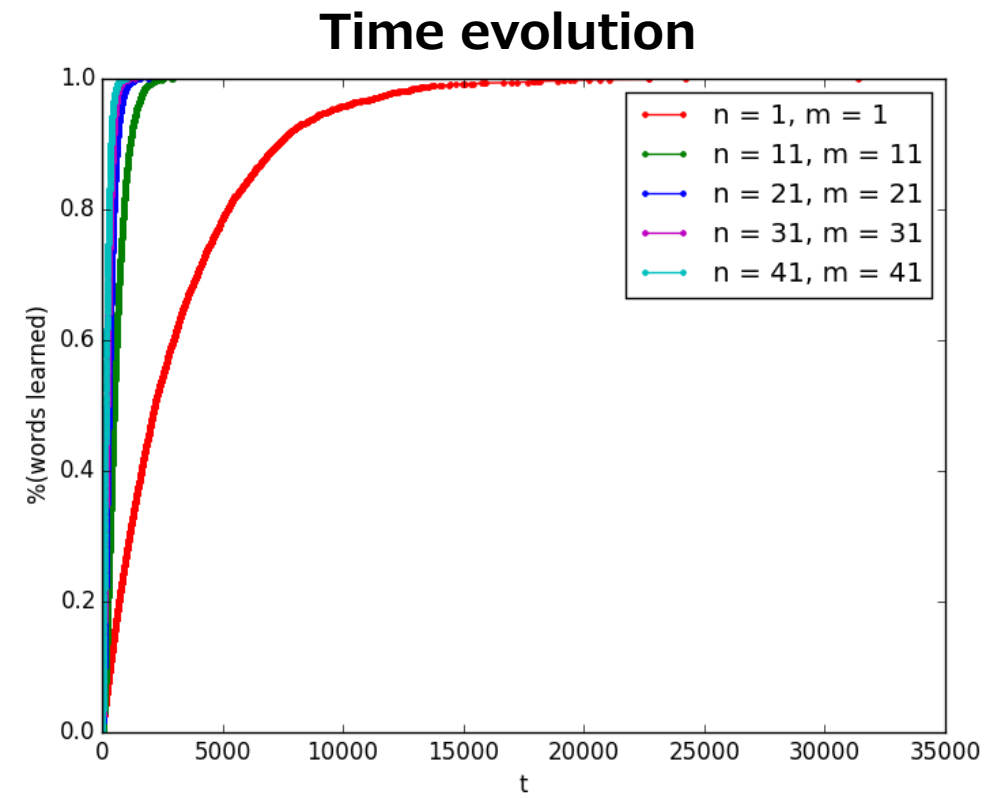
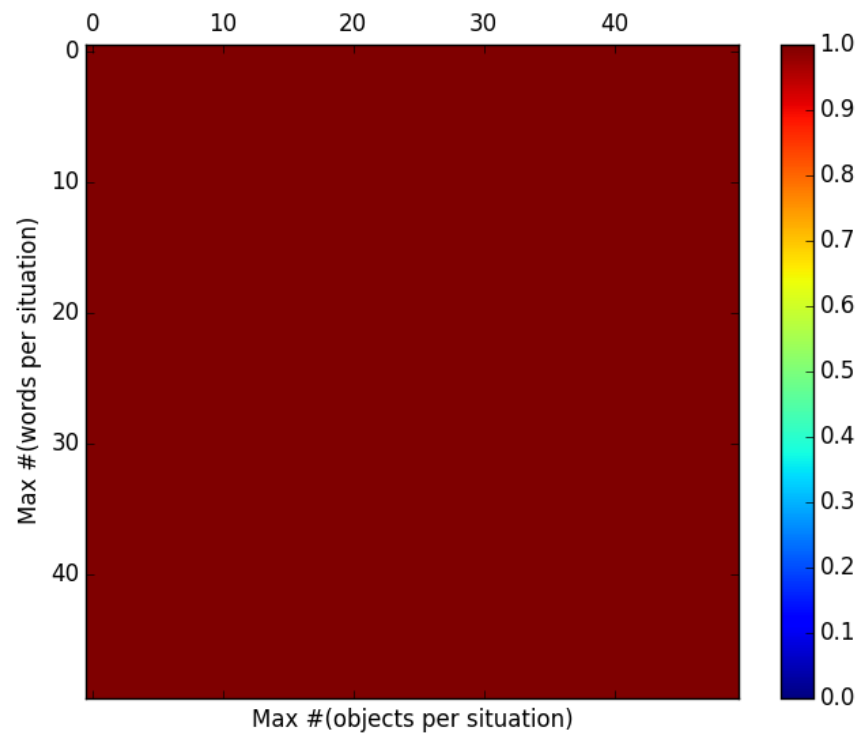
SUN



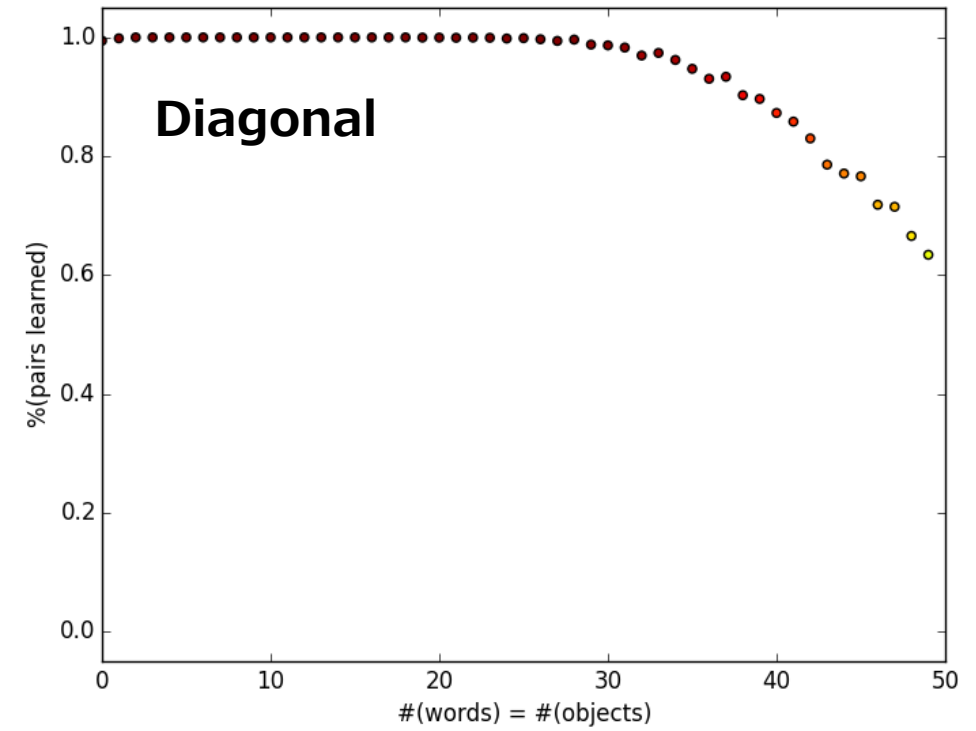
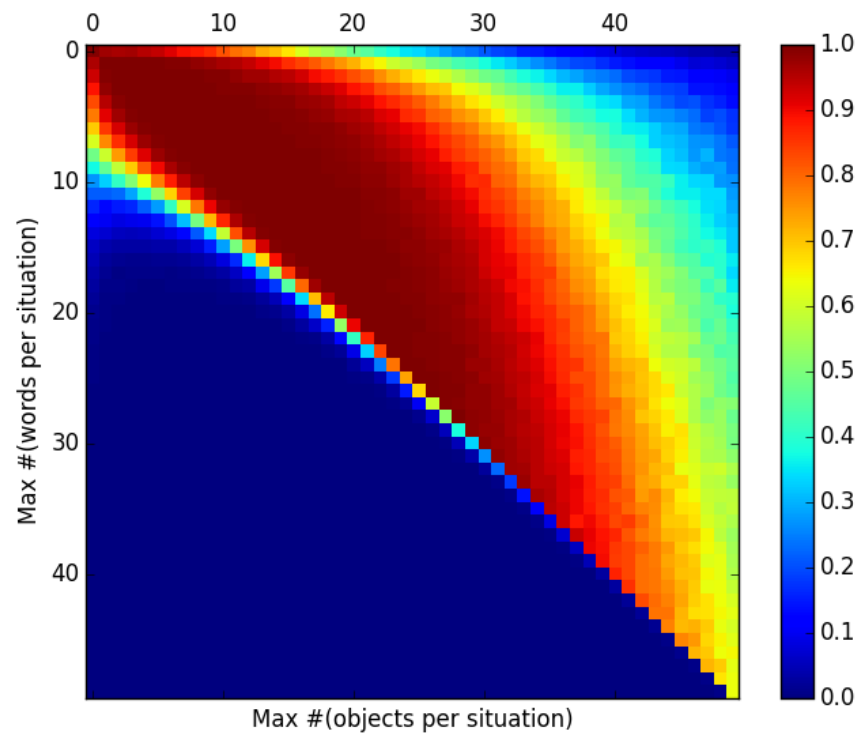
Uniform



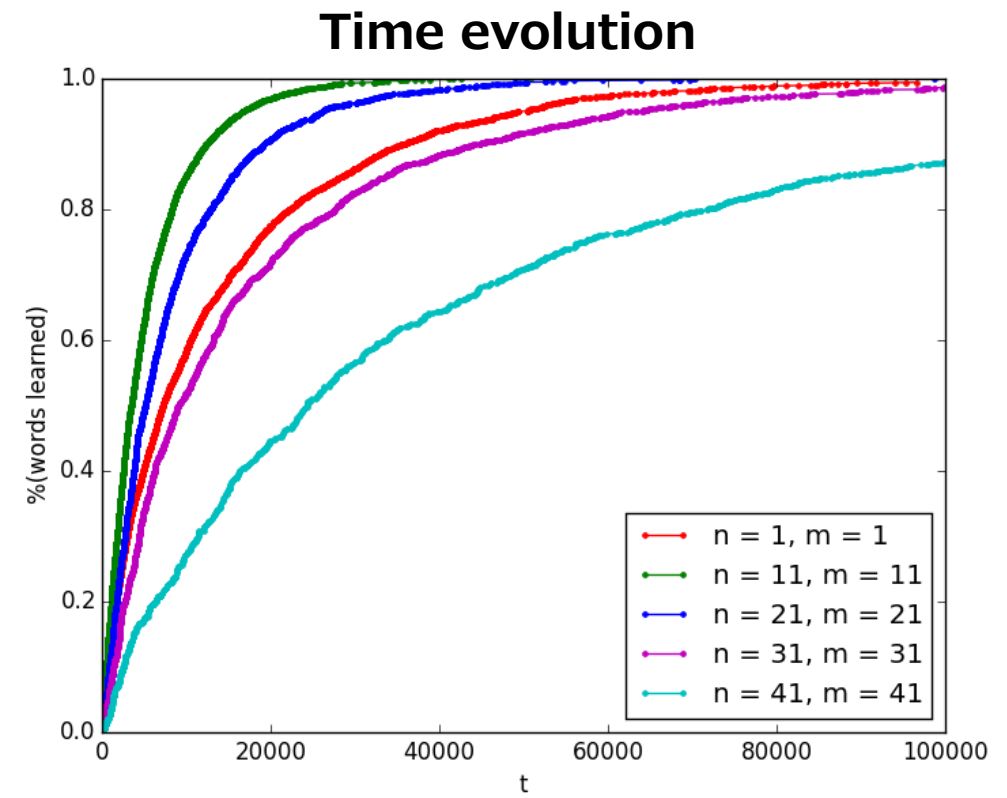
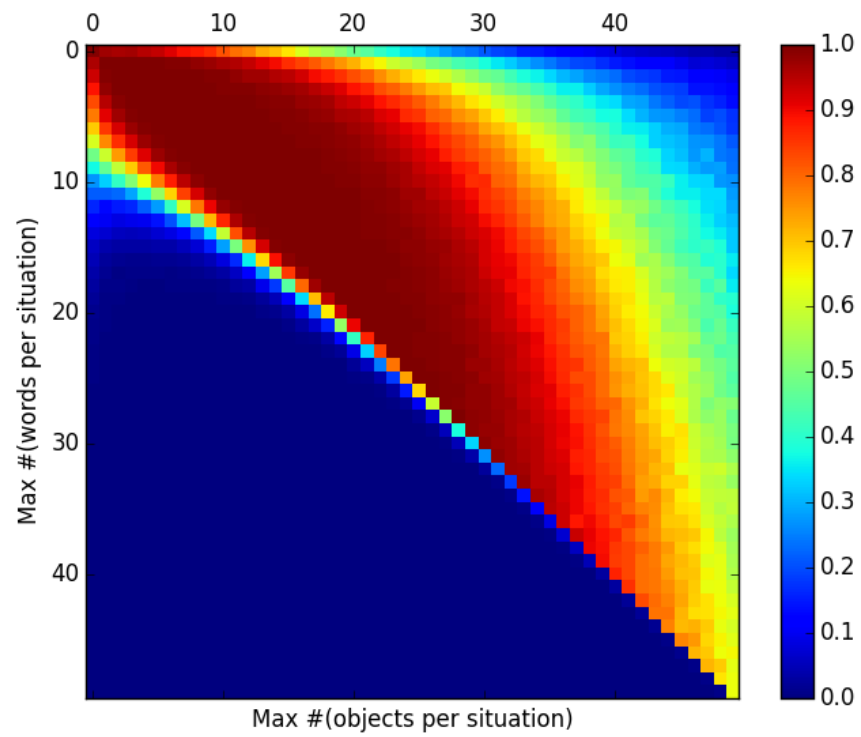
Uniform



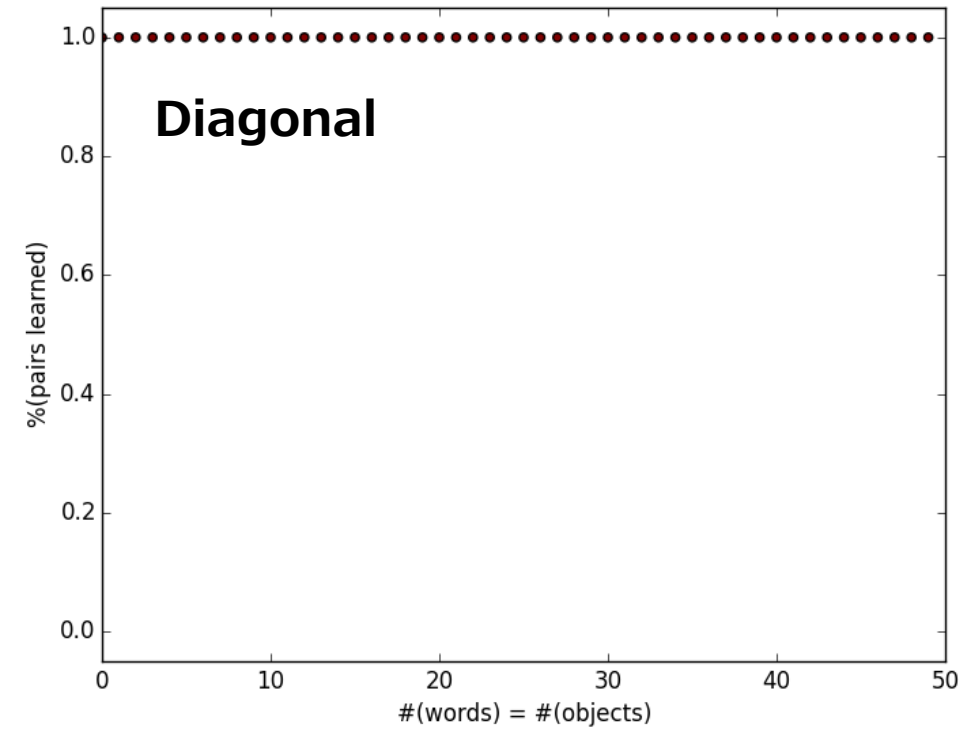
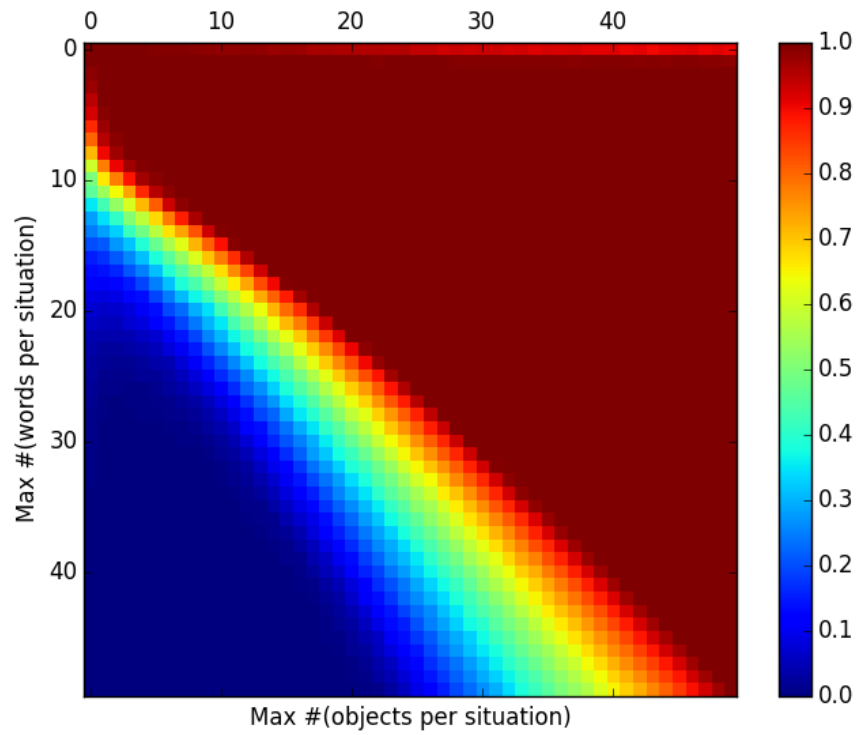
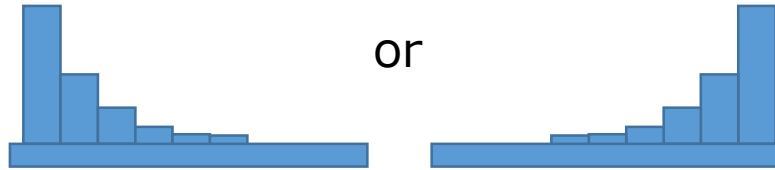
Zipf



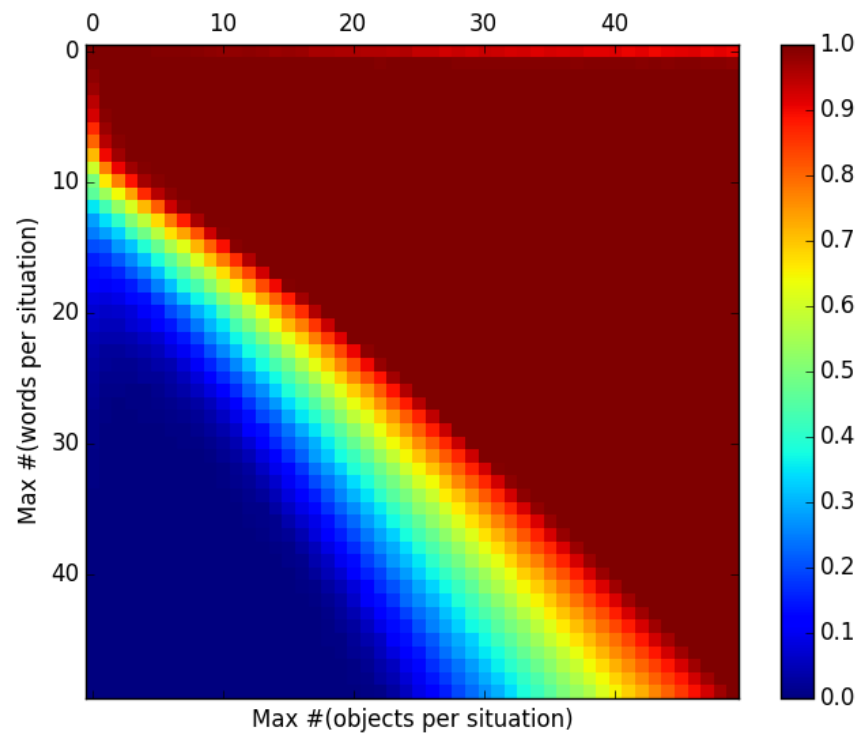
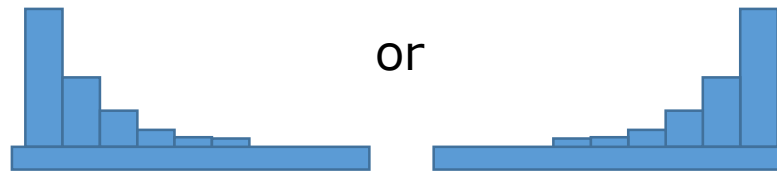
Zipf



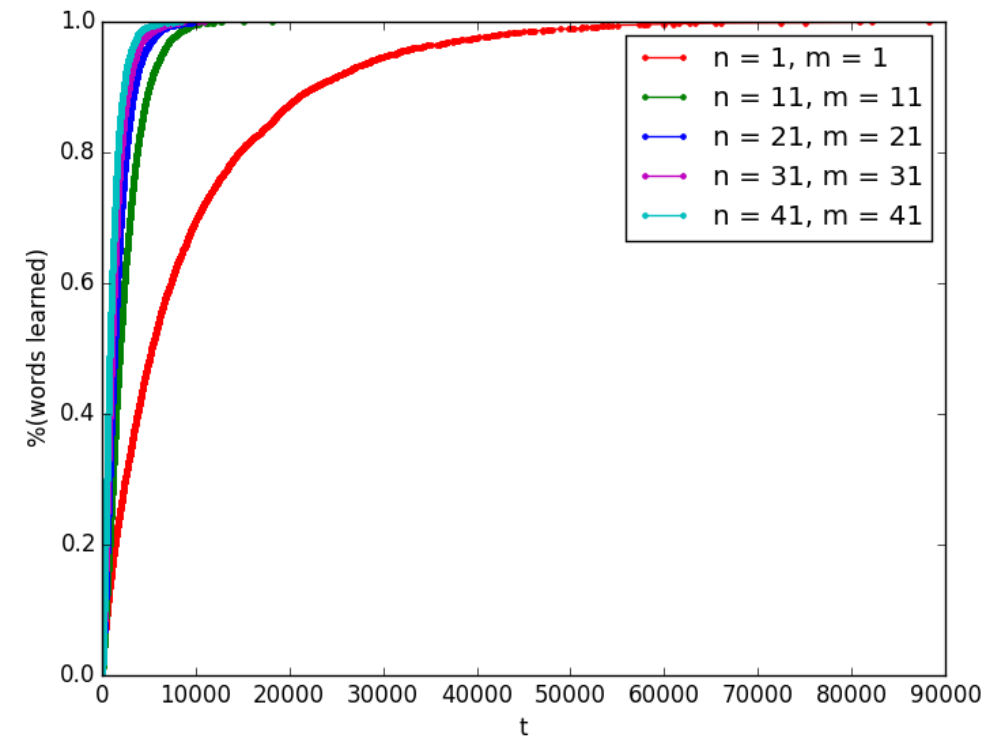
Two Zipf



Two Zipf



Time evolution



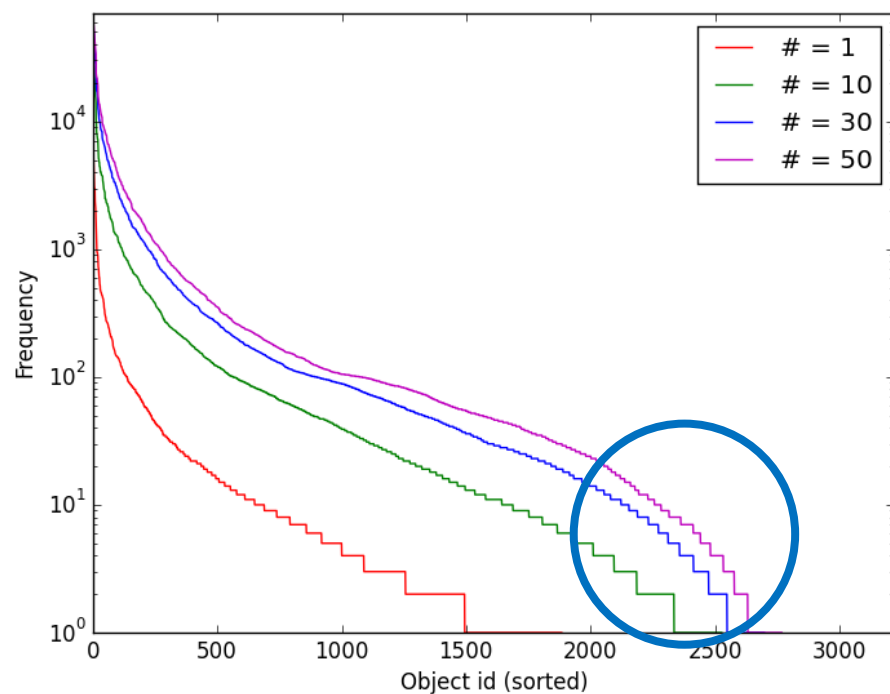
Summary & Discussion

- When Uniform, the learner got all 3200 words
- SUN and Zipf are resemble
 - Best performance if $\#words = \#objects$
 - Better if $\#words < \#objects$
 - Worse if $\#words > \#objects$
 - Peak is near $\#words = \#objects = 10$??Why??
- But When Zipf, the learner got more words
 - Using SUN gets worse! ??Why??

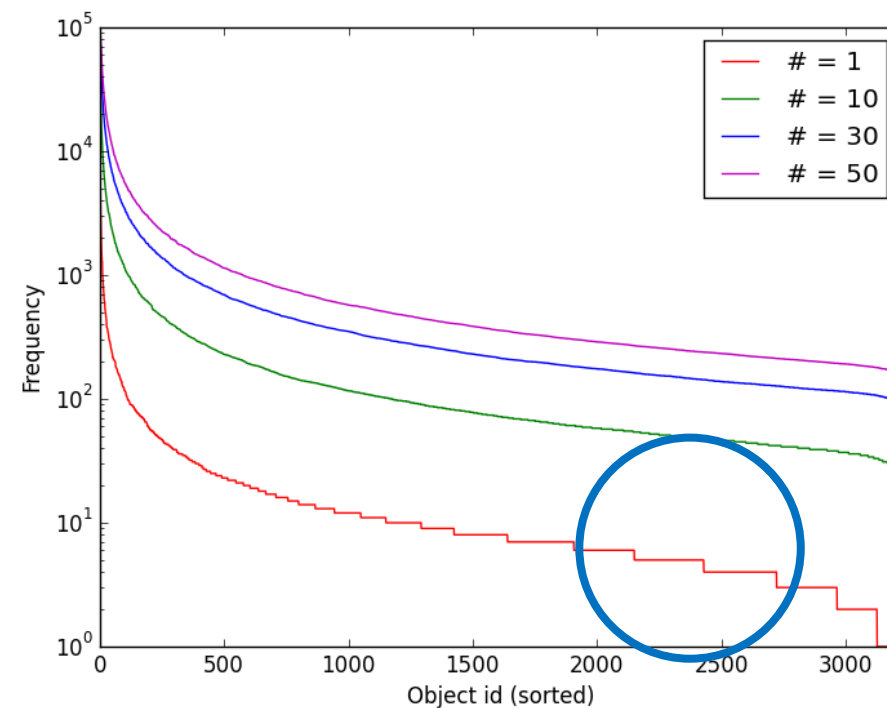
Frequency in learning dataset

SUN has low-frequency objects more than Zipf
Is this why Using SUN gets worse?

SUN



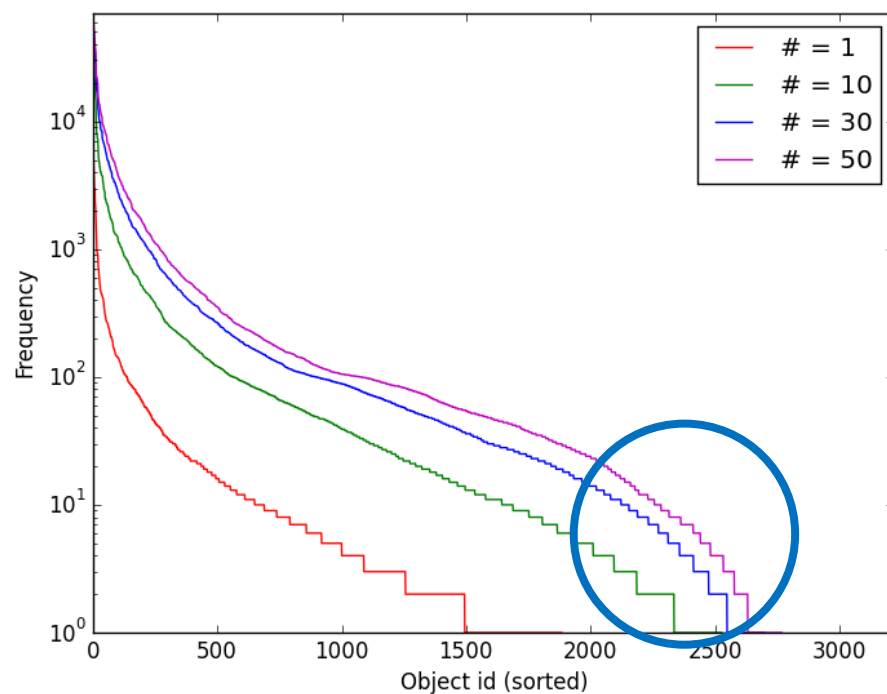
Zipf



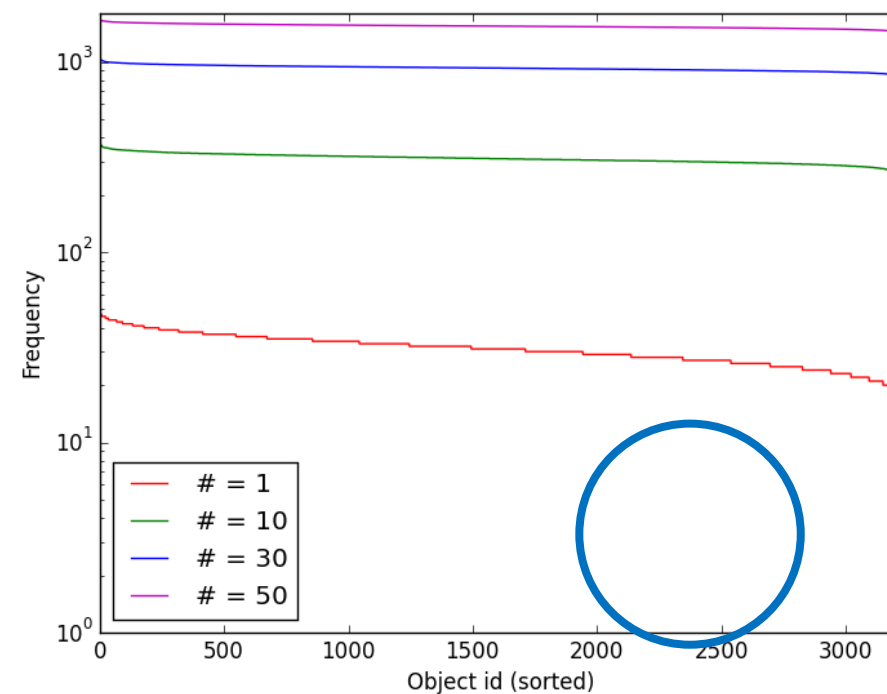
Frequency in learning dataset

SUN has low-frequency objects more than Uniform
Is this why Using SUN gets worse?

SUN



Uniform



Simulation 2 (coin flip)

20160603

Simulation 2 (shohei procedure)

The SUN database, row probability matrix, $P = (p_{ij})$.

Each p_{ij} of object j of scene i is binomial.

Assume objects = words in the database.

Introducing a parameter a , using a new matrix aP , the expected number of words is N .

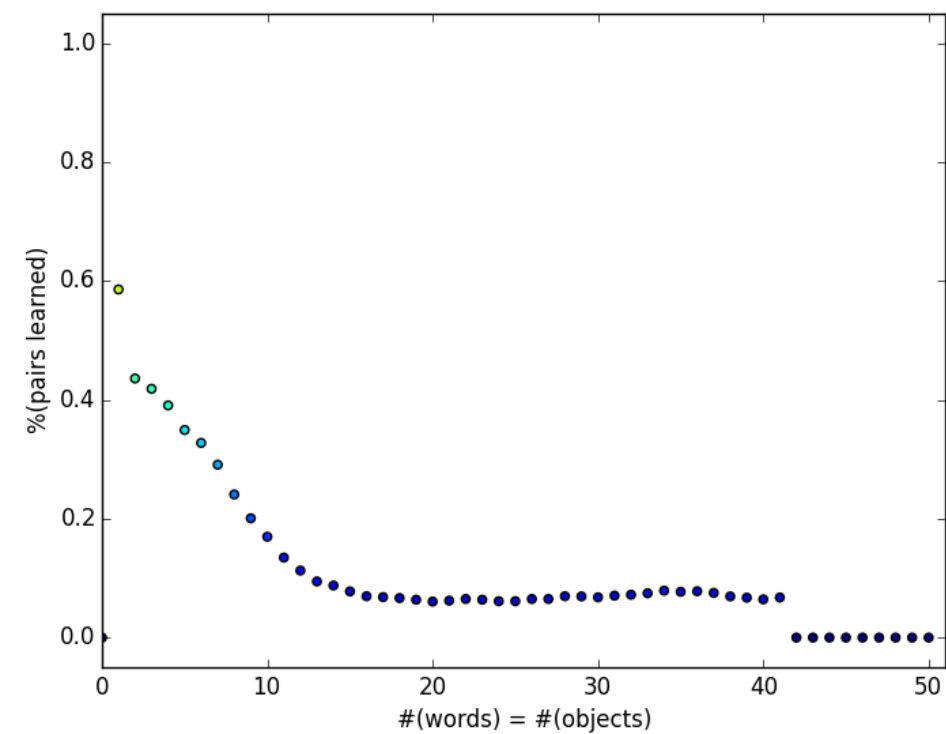
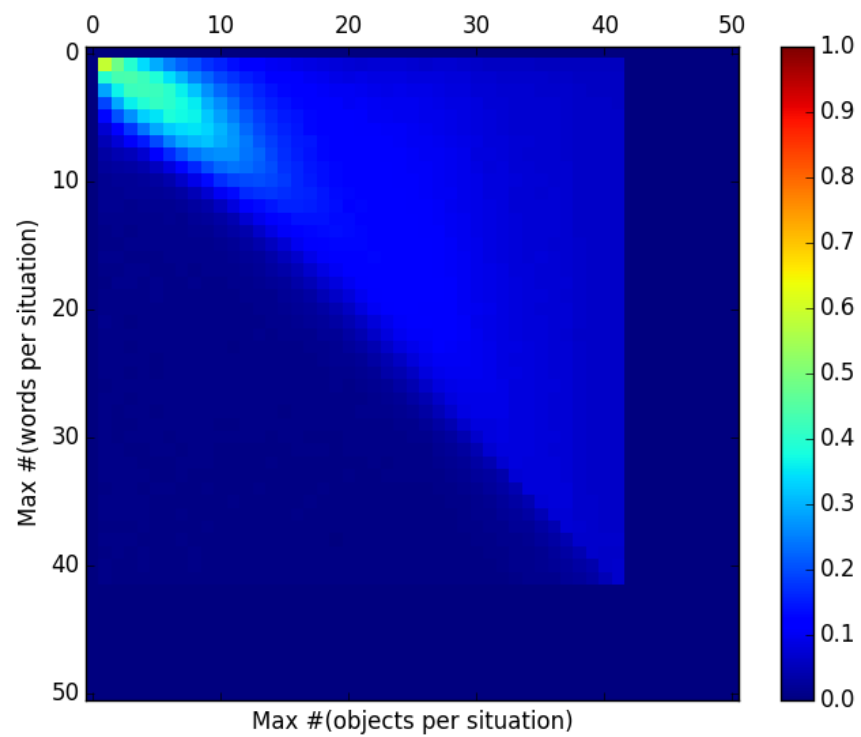
Likewise, a parameter b , a new matrix bP , the expected number of objects is M .

Suppose “always containing the answers”.

Using bP to determine only the number of objects, but the actual contents are replaced by the subset corresponding to the words.

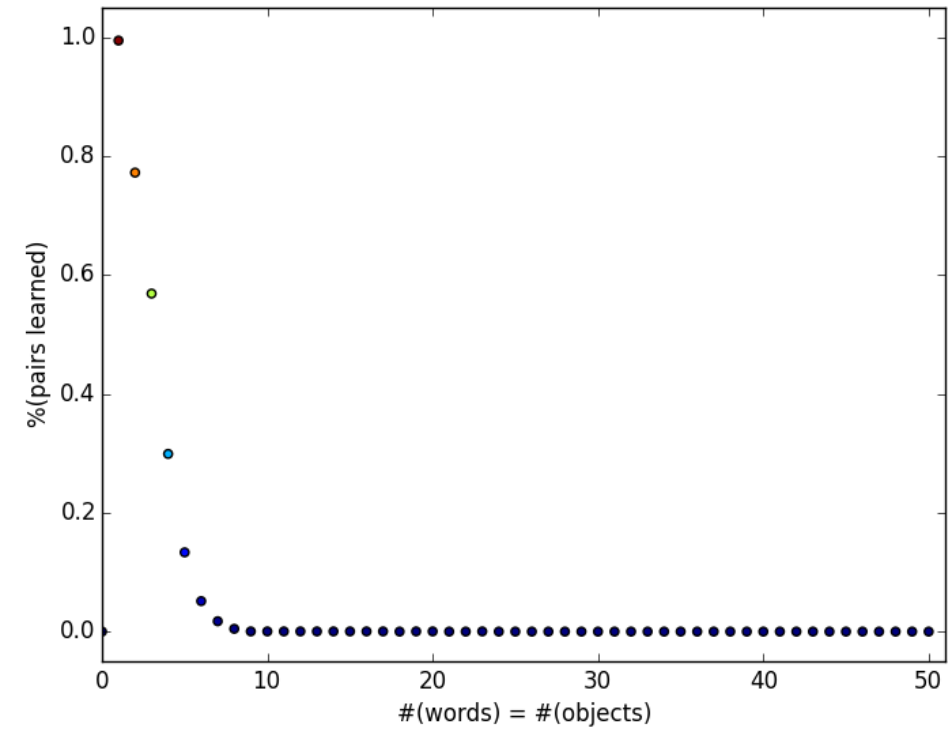
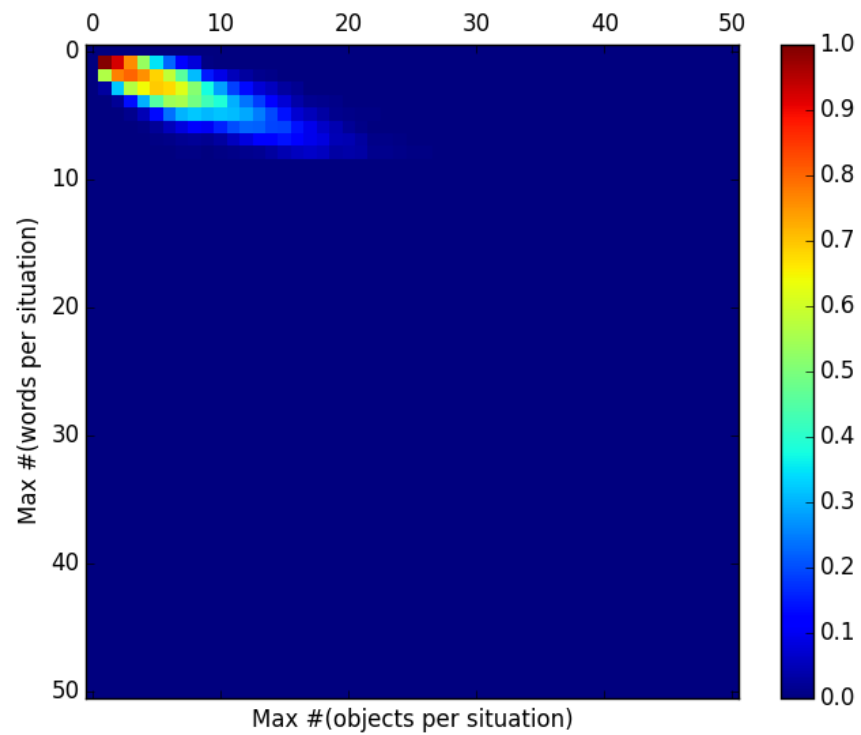
In average, the number of words is N , the number of objects is M .

SUN



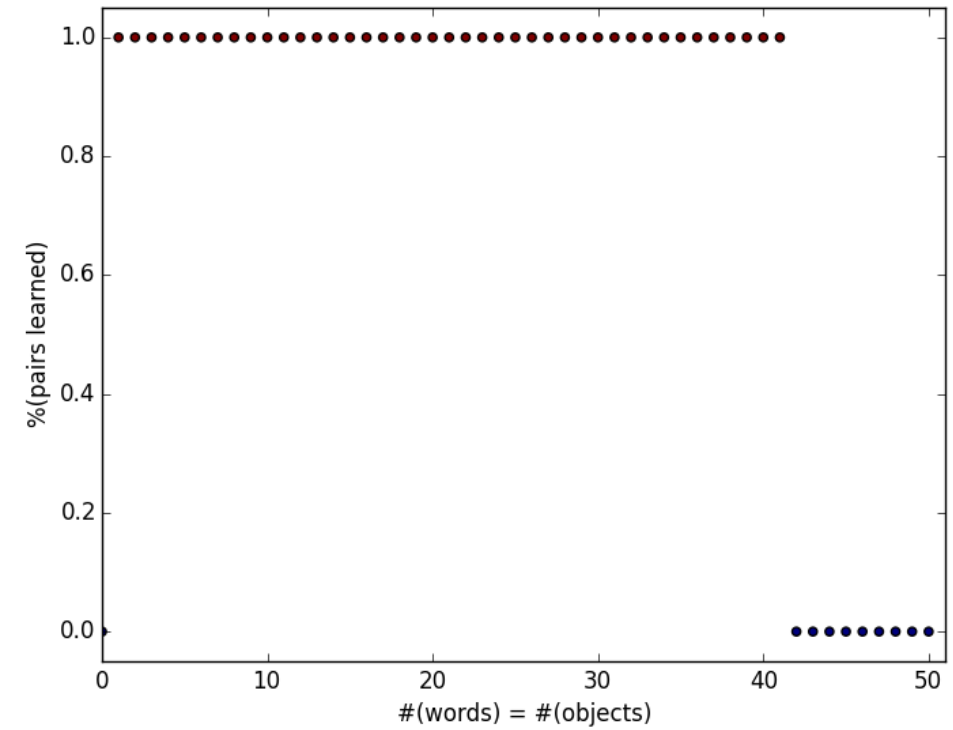
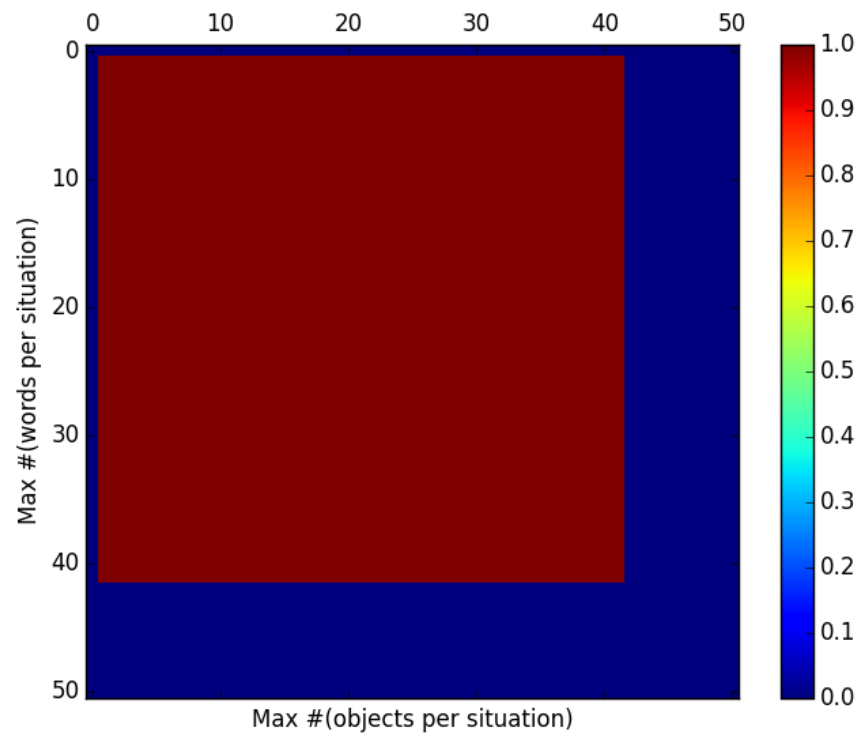
VERY HARD to learn

Zipf



VERY HARD to learn

Uniform

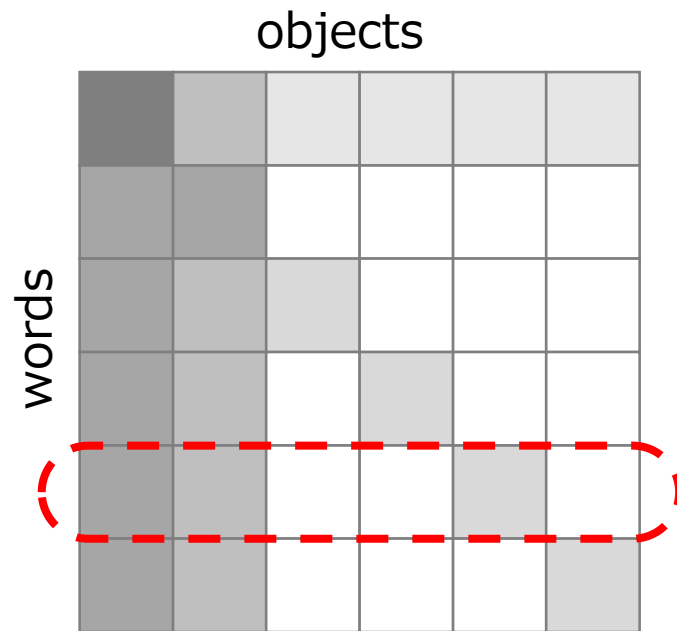


Simulation 3 (coin flip + mutex bias)

20160803

Background: The barrier of learning

- The anti behavior to the learning condition

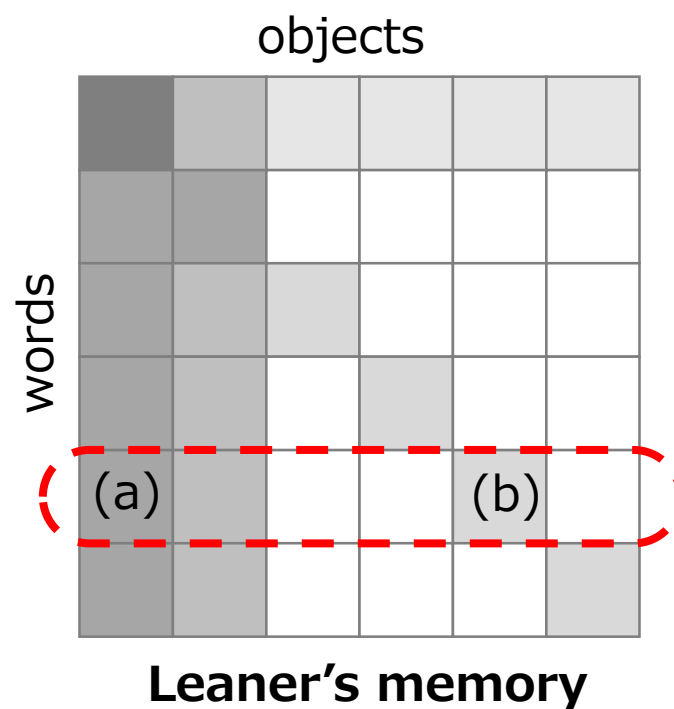
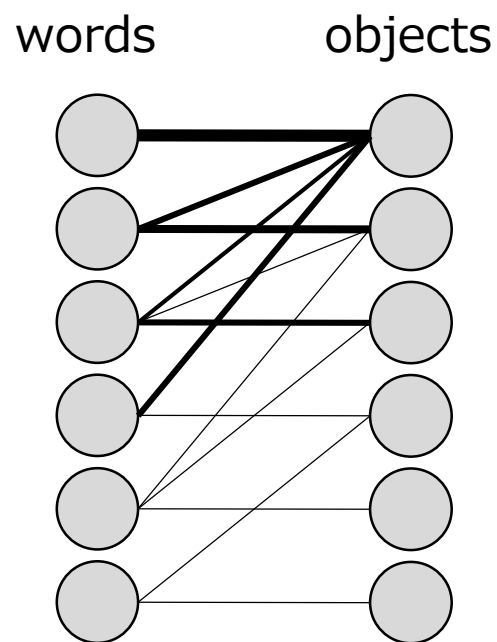


In every row, $F(1st) > F(diag)$

Counter to the learning condition:
learned if $\max F(offdiag) < F(diag)$

Mutual exclusivity bias

- Use this mutex bias for evaluation
- No change in their learning process
- 1 object $\leftrightarrow$ 1 word
- Weighted bipartite matching: Hungarian algorithm
 - Find maximum weight 1-to-1 pairs



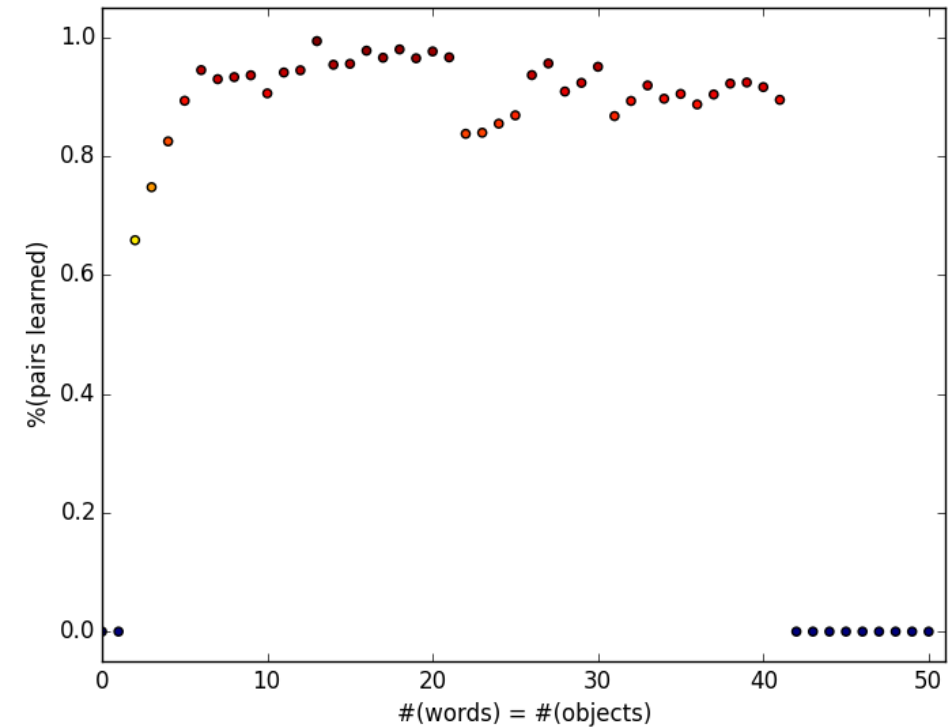
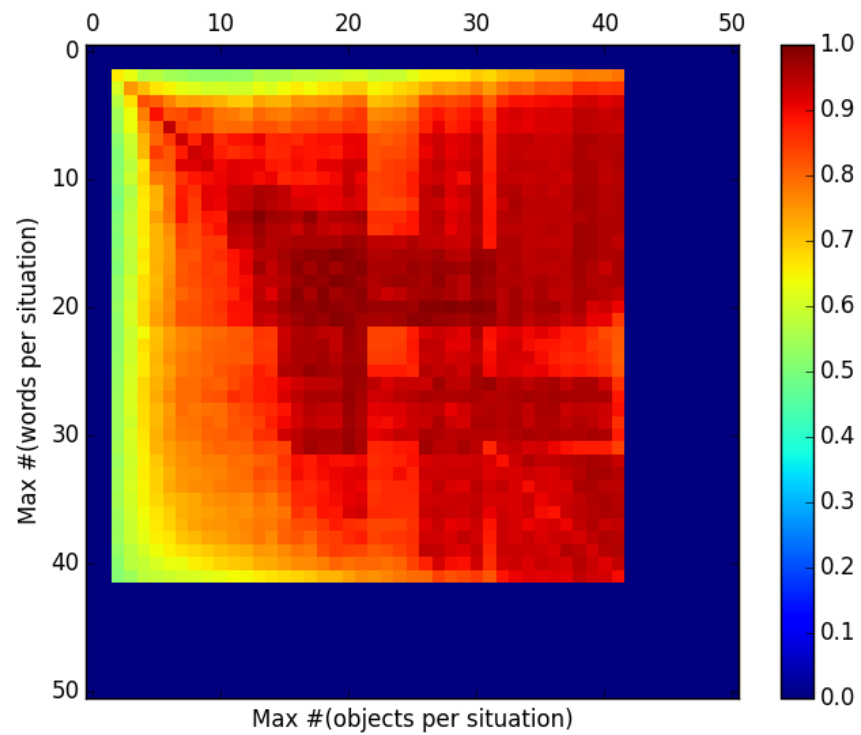
Frequency approach

$F(a) > F(b)$ so Word 5 $\rightarrow$ Object 1
Learning failure

Mutex approach most likely

(a) Word 1 $\rightarrow$ Object 1
so (b) Word 5 $\rightarrow$ Object 5

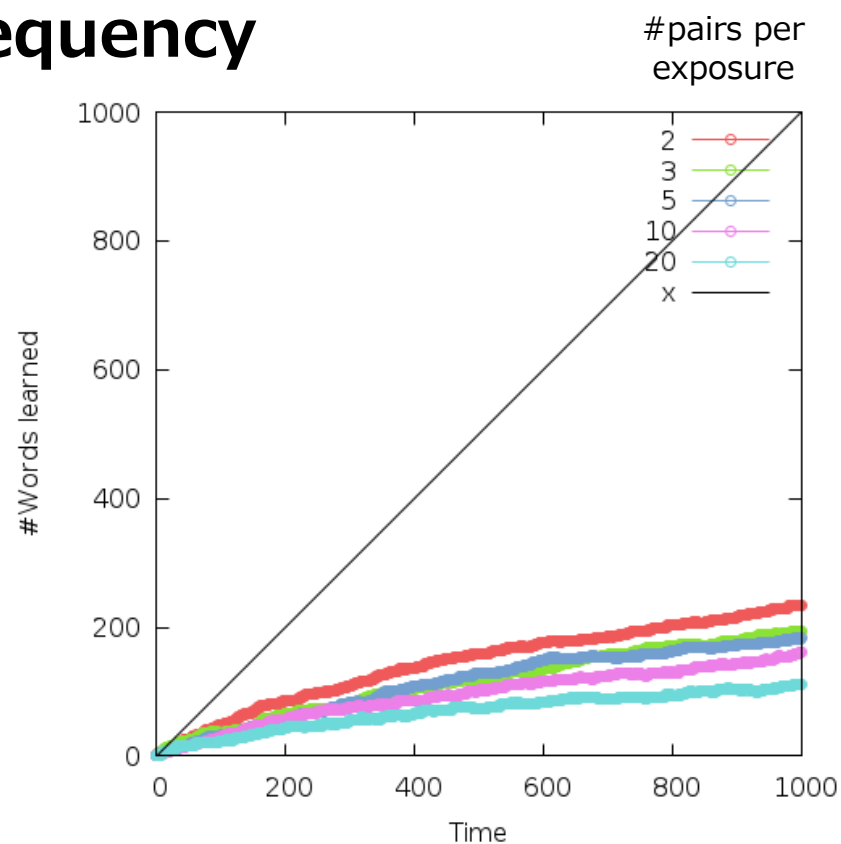
SUN (mutex)



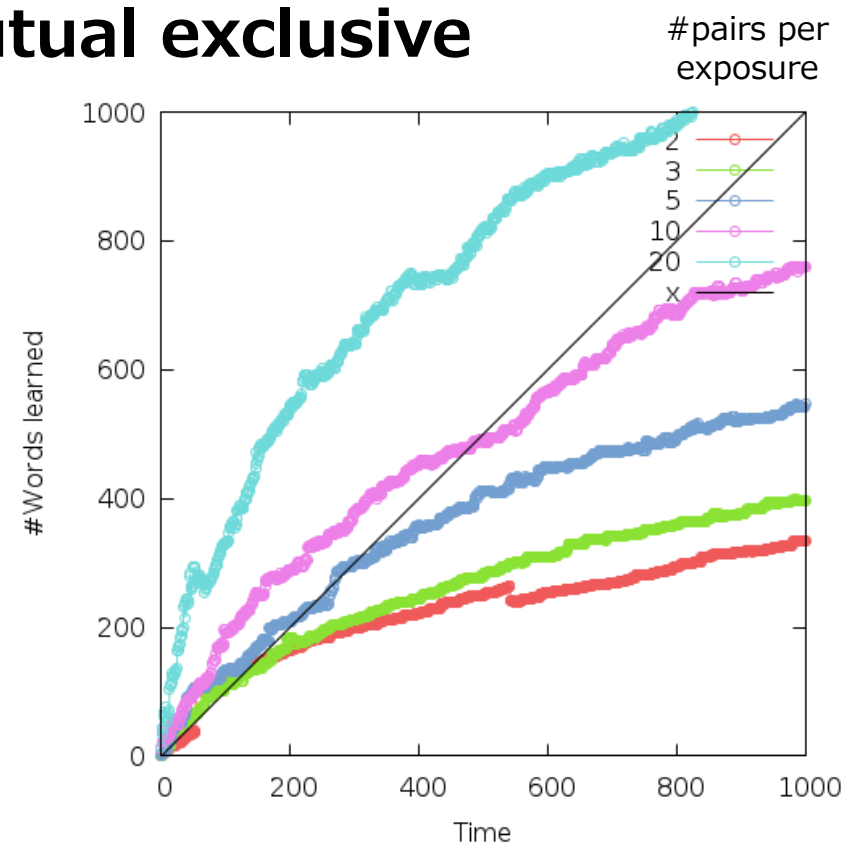
Asymmetry + Strange patterns

SUN: Learning speed (1-1000 step)

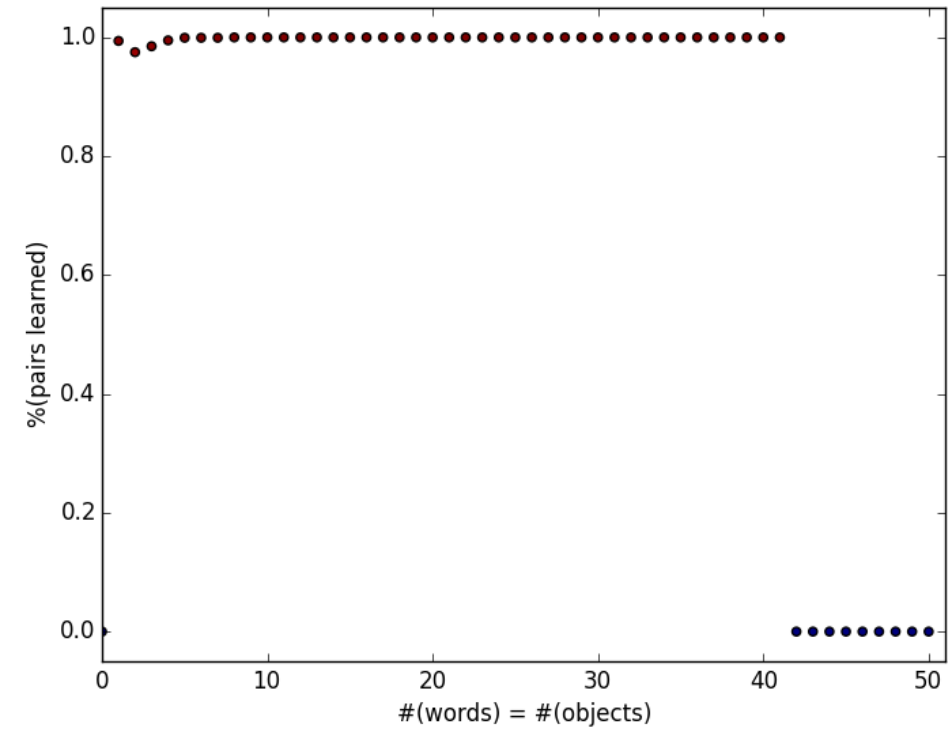
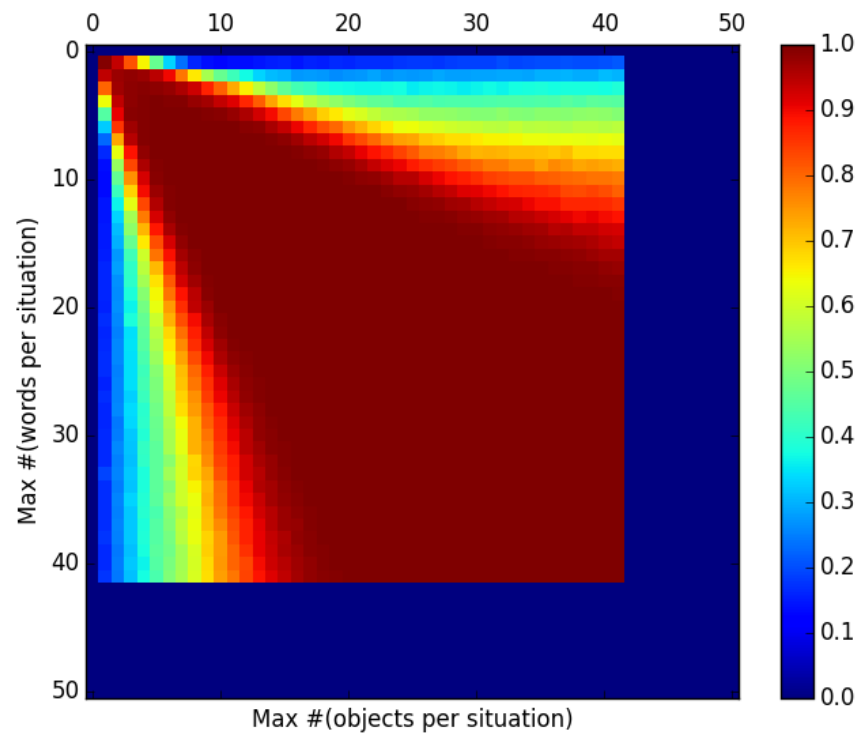
Frequency



Mutual exclusive



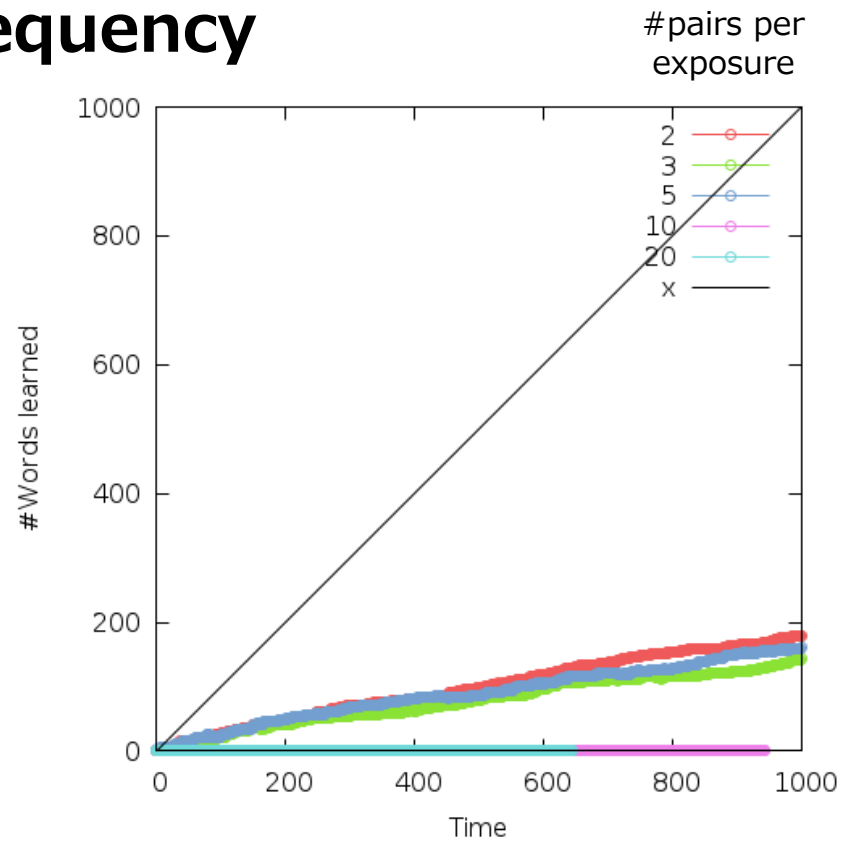
Zipf (mutex)



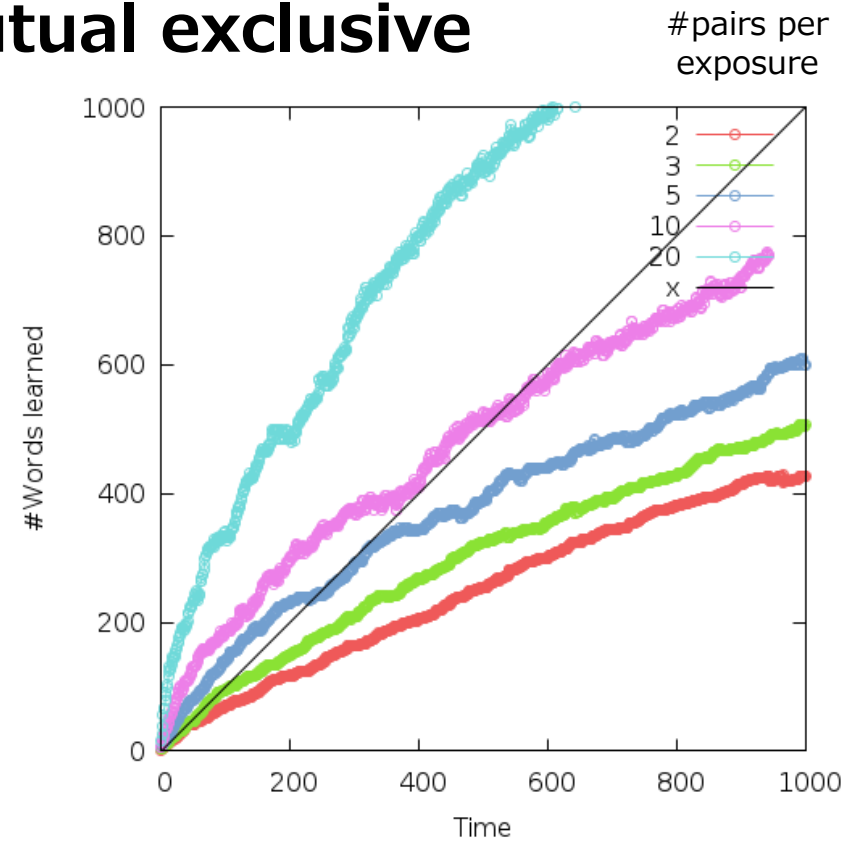
No Asymmetry

Zipf: Learning speed (1-1000 step)

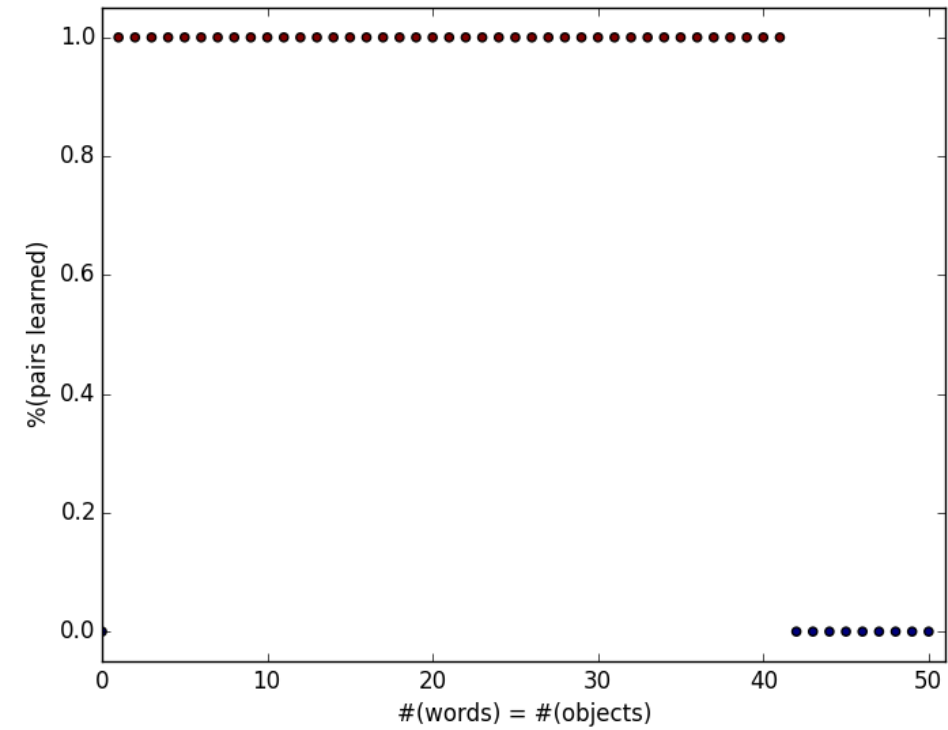
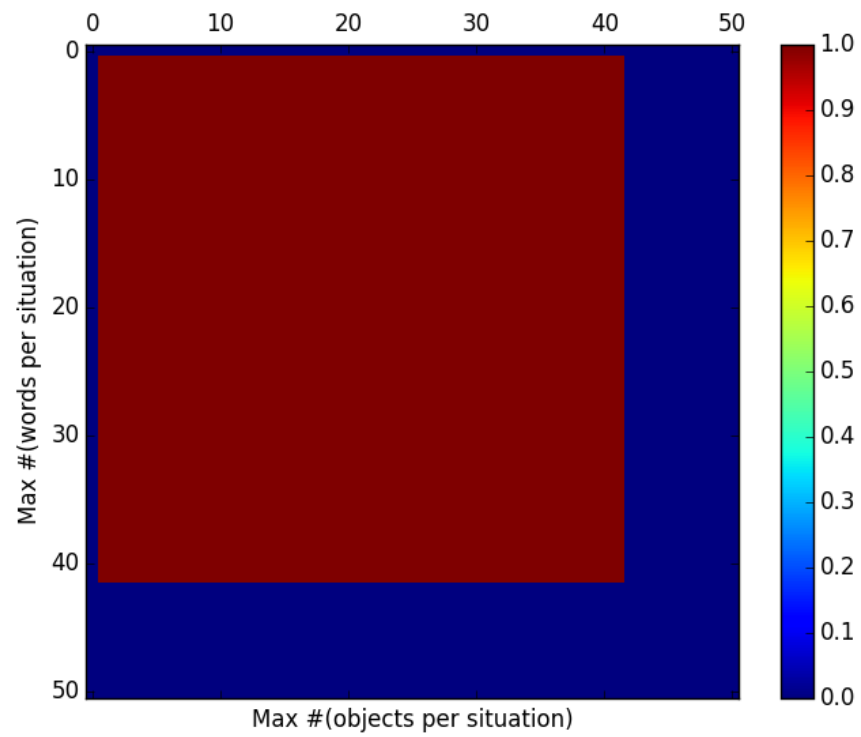
Frequency



Mutual exclusive

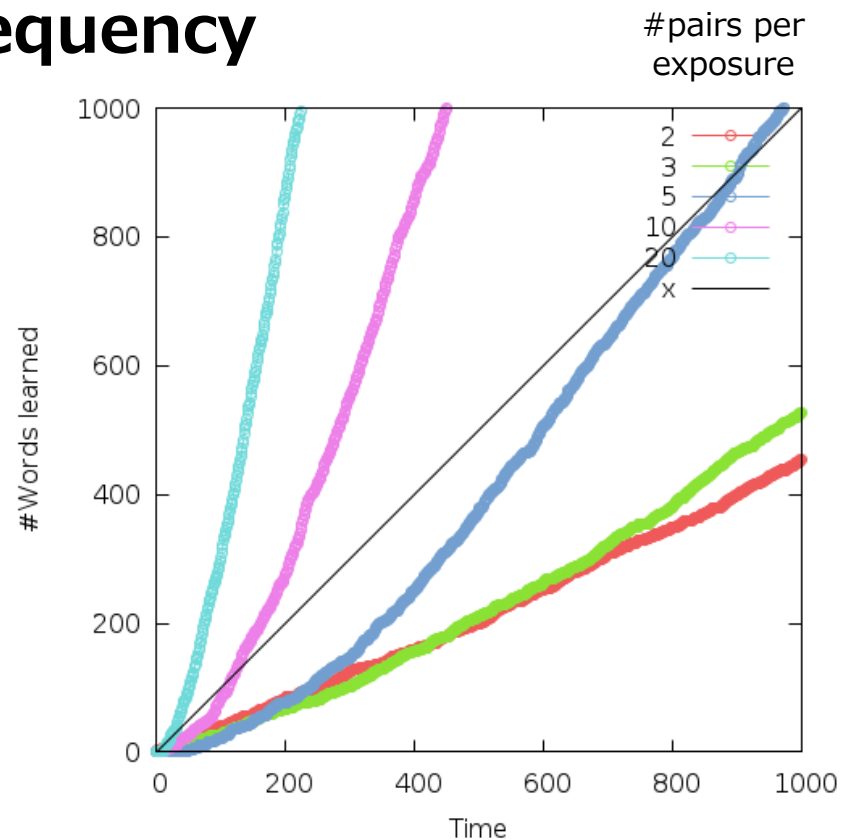


Uniform (mutex)

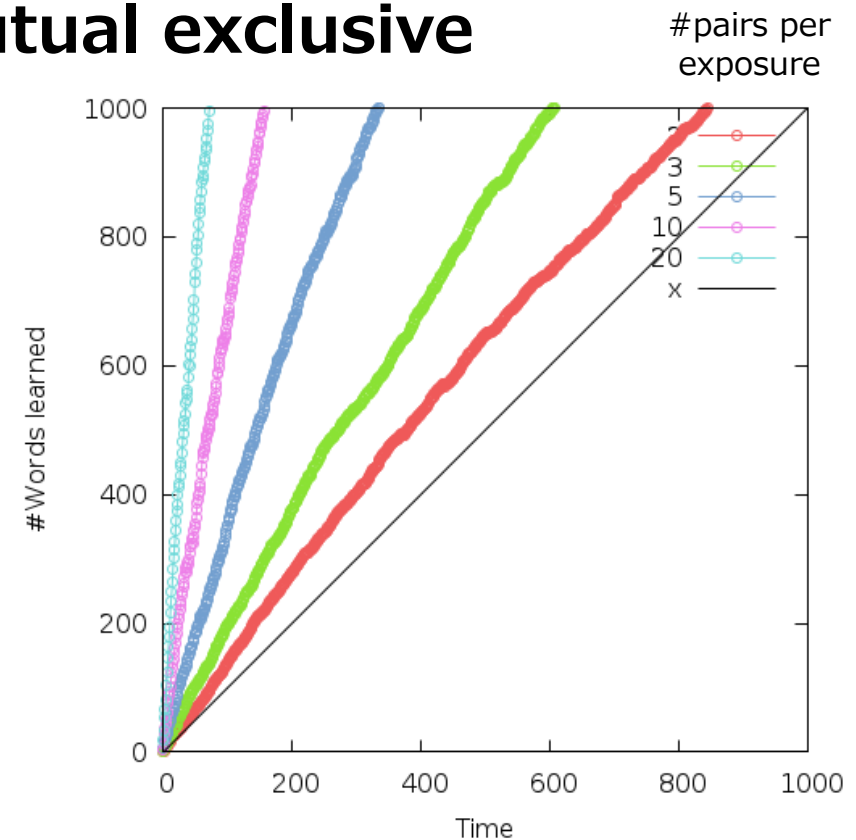


Uniform: Learning speed (1-1000 step)

Frequency



Mutual exclusive

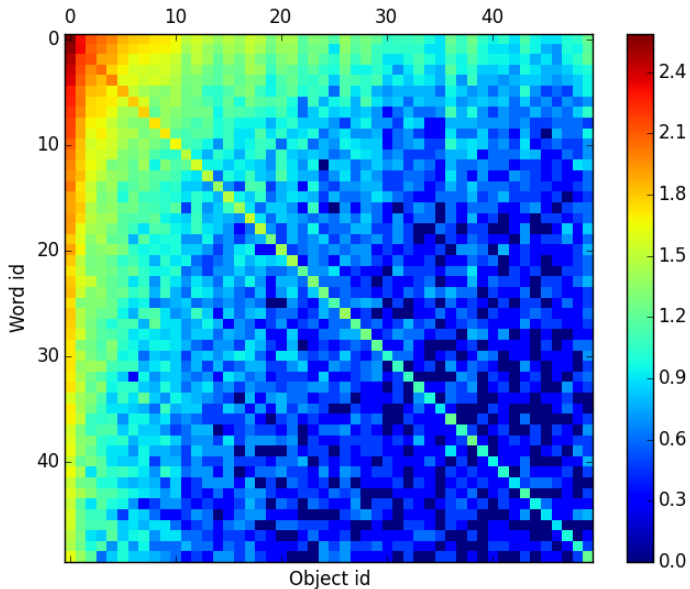


Learner's memory in simple cases

Zipf: $p=50$, $T=100$

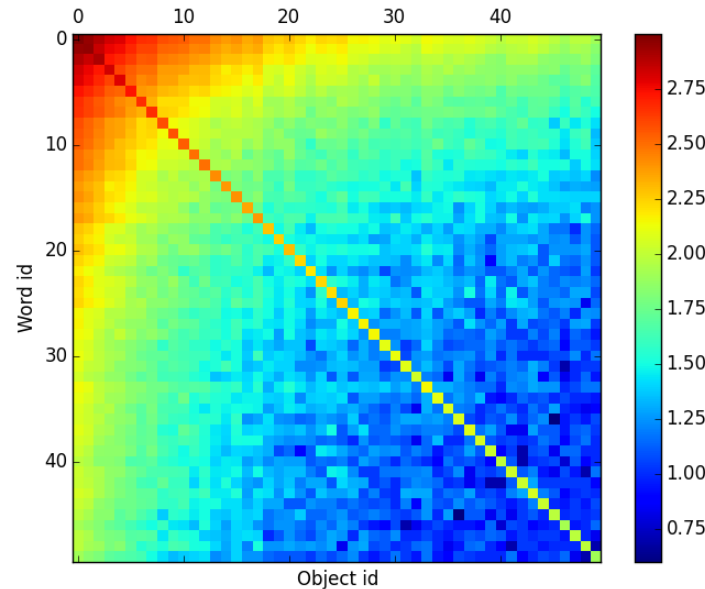
In each situation

$\#words = 20$, $\#obj = 2$

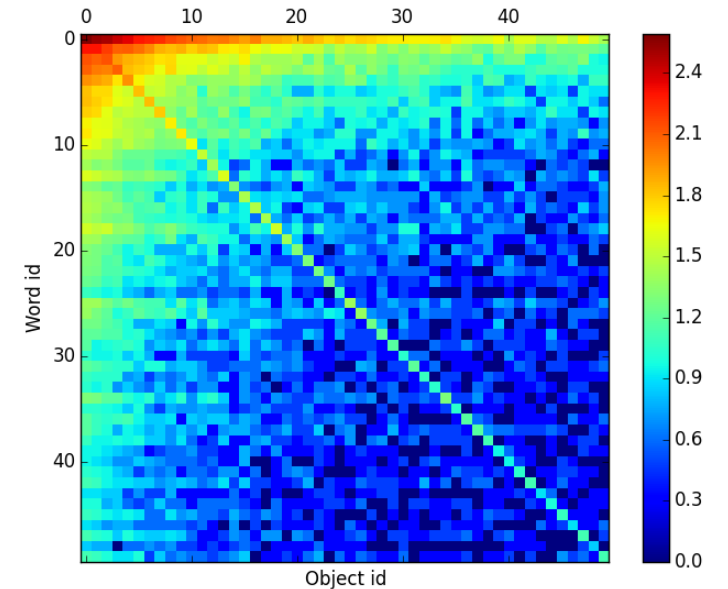


$\#words > \#obj \rightarrow$ vertical stripes
1st object prevents learning

$\#words = 20$, $\#obj = 20$



$\#words = 2$, $\#obj = 20$

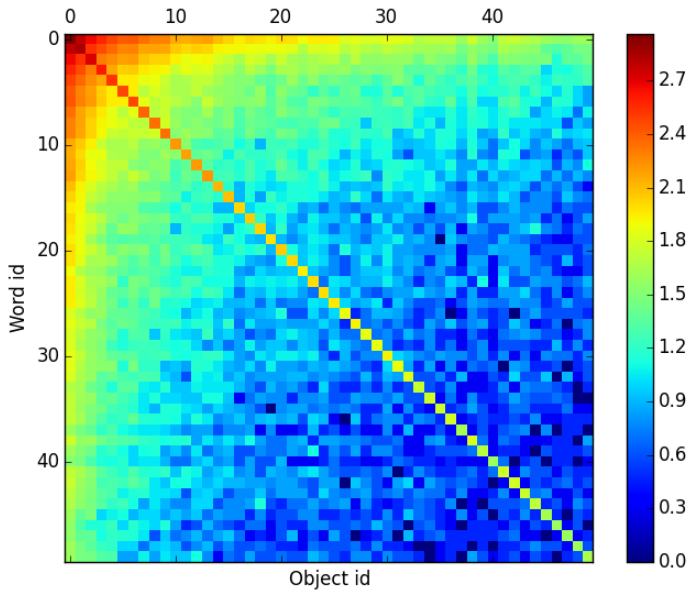


$\#words < \#obj \rightarrow$ horiz stripes
No one prevents learning

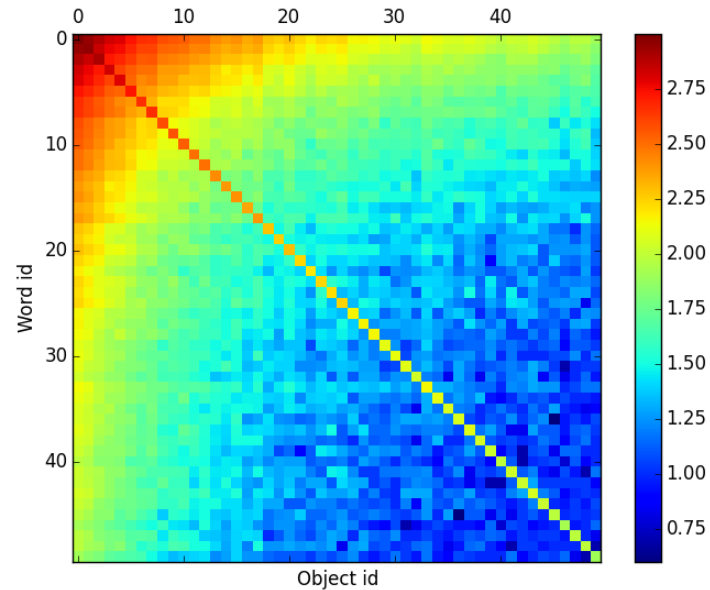
Zipf: $p=50$, $T=100$

In each situation

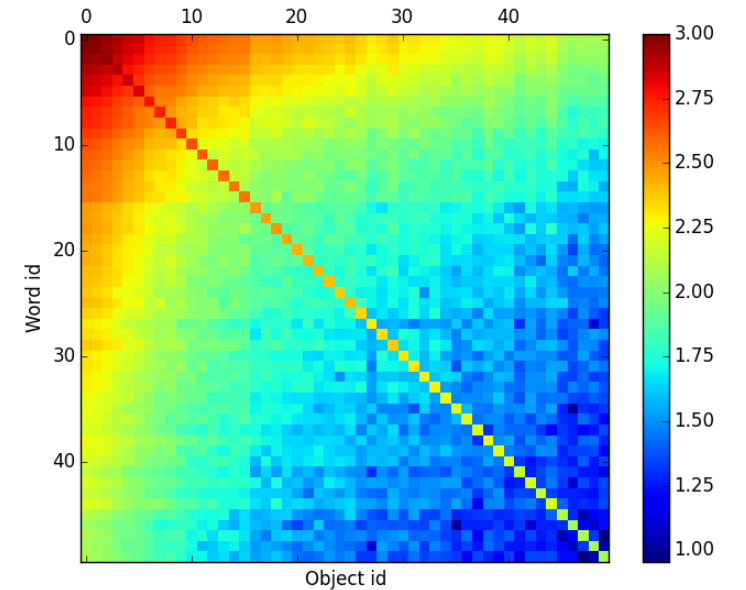
#words = 10, #obj = 10



#words = 20, #obj = 20



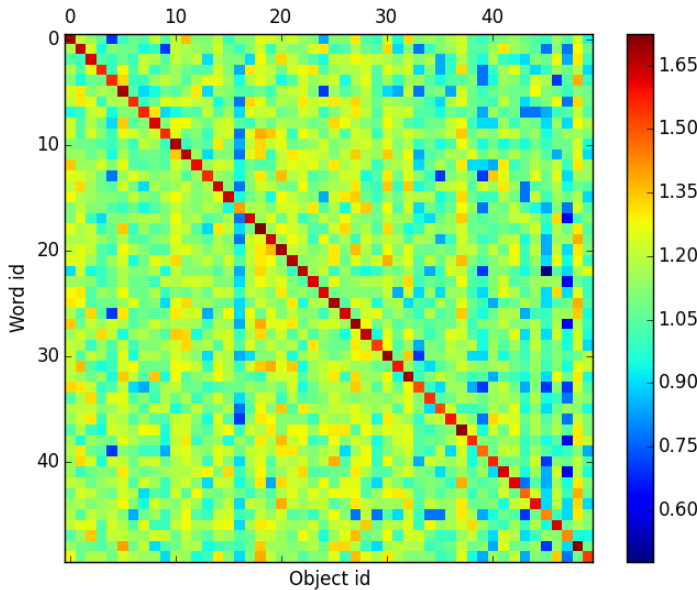
#words = 30, #obj = 30



Uniform: $p = 50, T = 100$

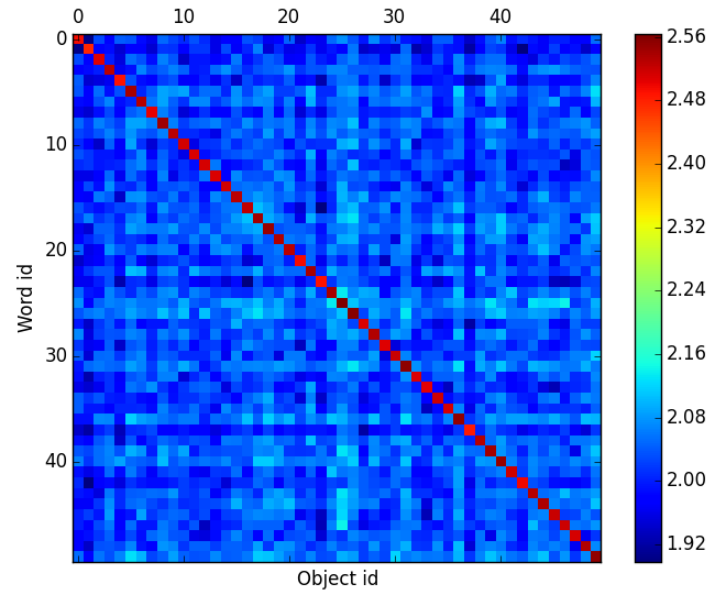
In each situation

$\#words = 20, \#obj = 2$

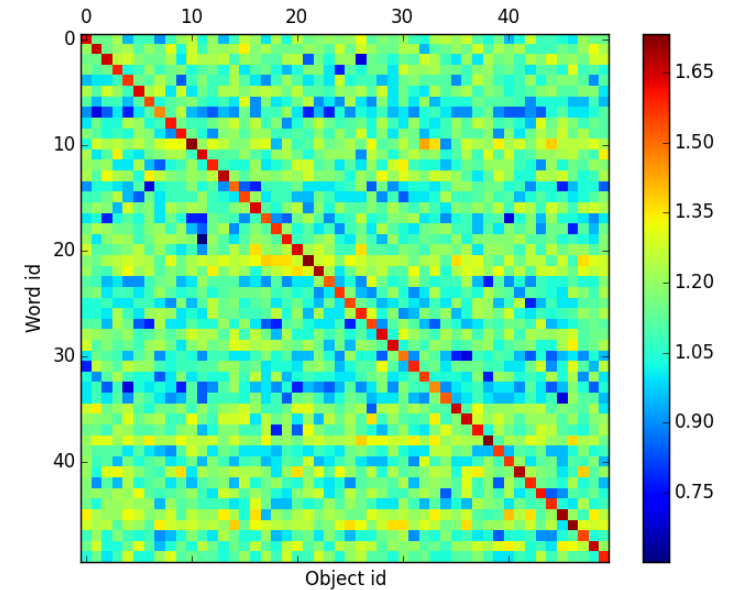


$\#words > \#obj \rightarrow$ vertical stripes

$\#words = 20, \#obj = 20$



$\#words = 2, \#obj = 20$

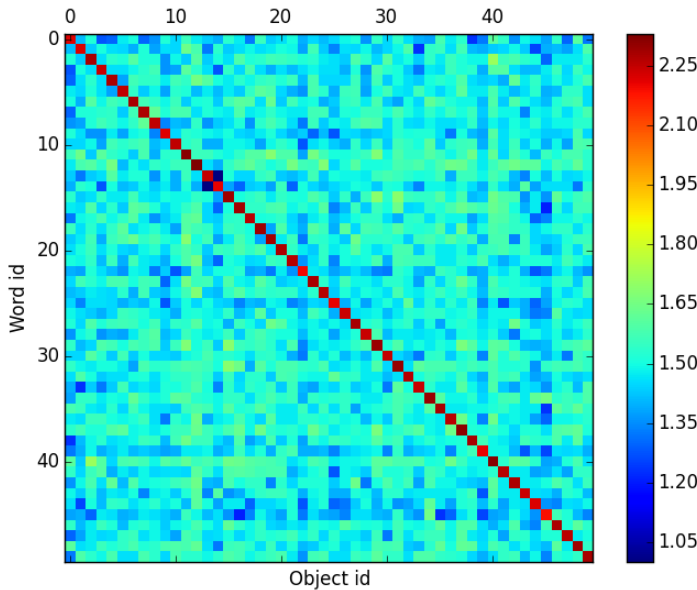


$\#words < \#obj \rightarrow$ horiz stripes

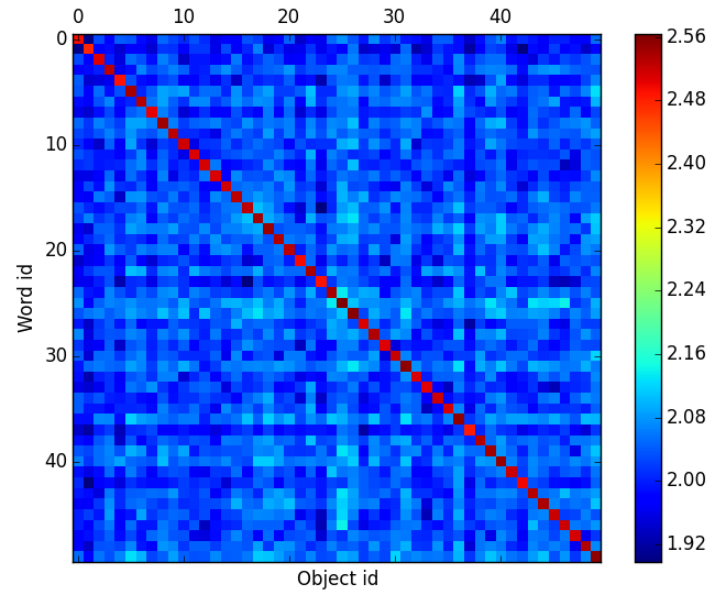
Uniform: $p = 50, T = 100$

In each situation

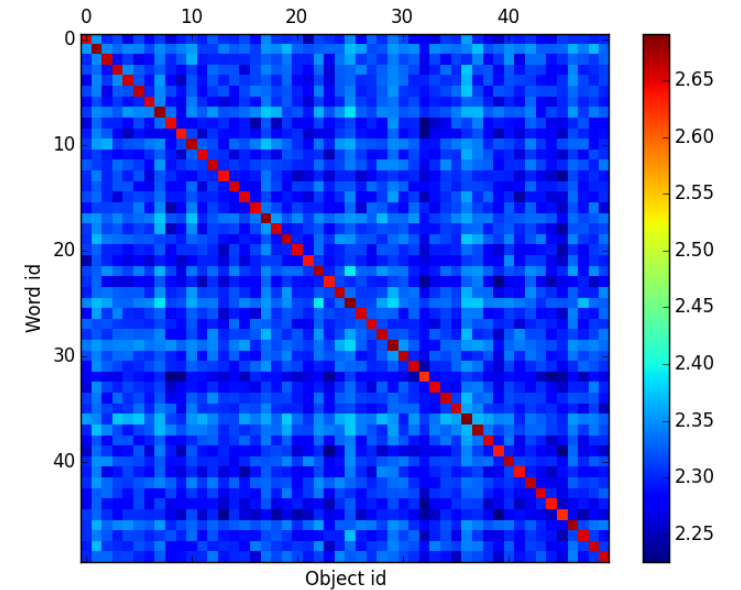
#words = 10, #obj = 10



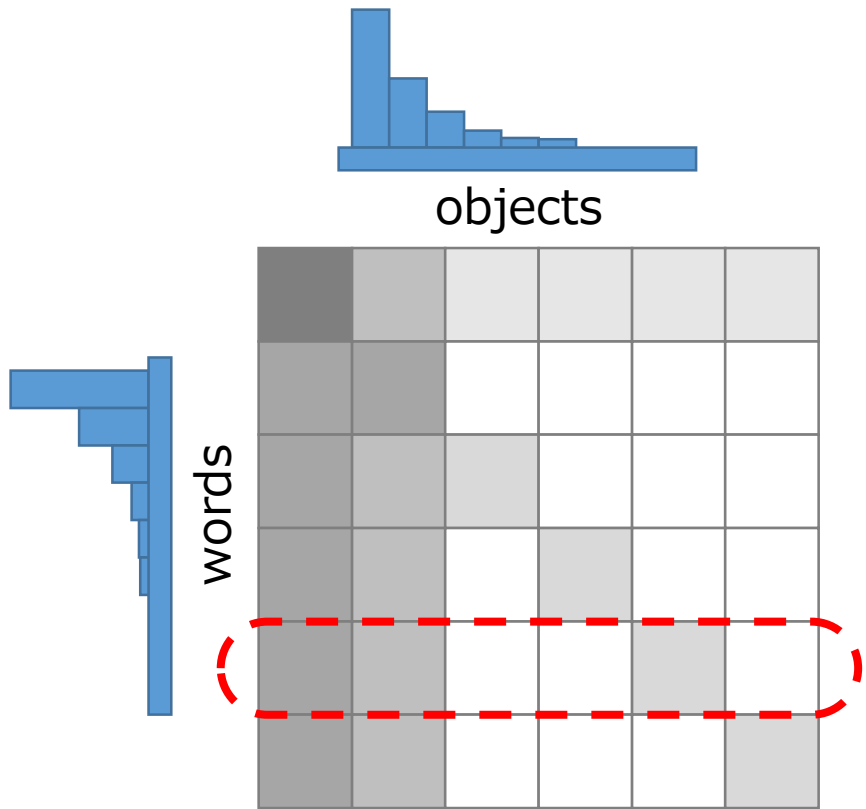
#words = 20, #obj = 20



#words = 30, #obj = 30



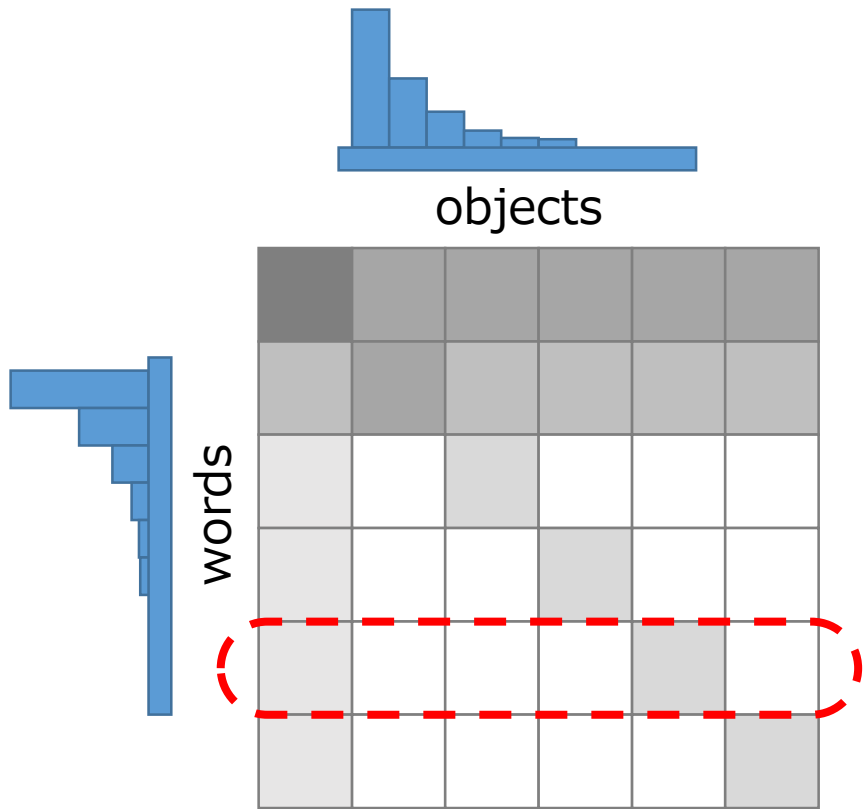
Schematic for Zipf: $\# \text{words} > \# \text{objects}$



In every row, $F(1st) > F(diag)$

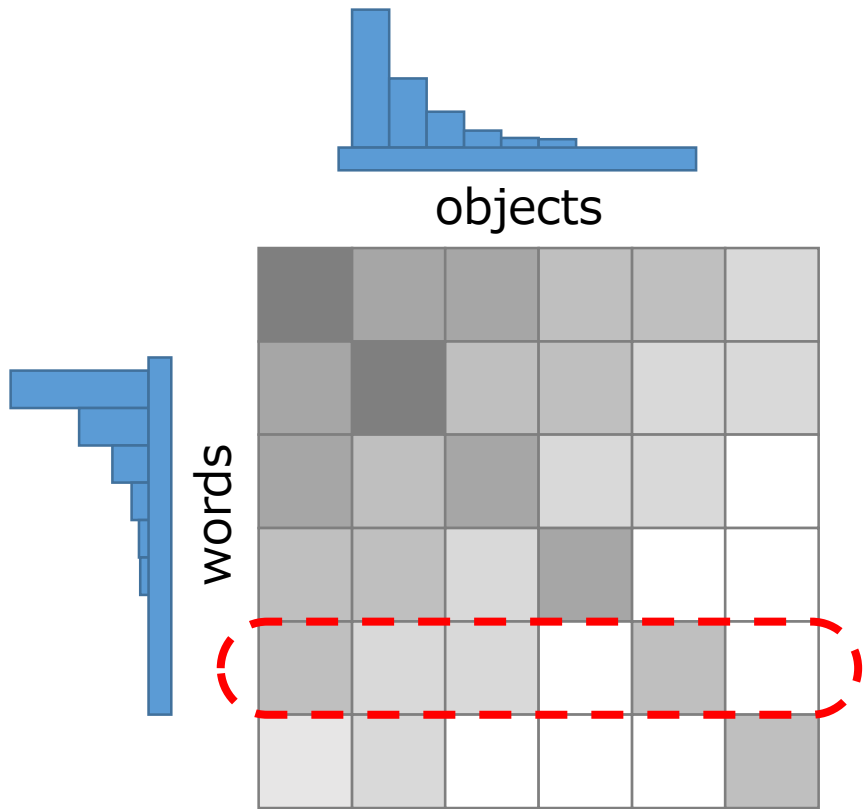
Counter to the learning condition:
learned if $\max F(offdiag) < F(diag)$

Schematic for Zipf: $\# \text{words} < \# \text{objects}$



Better to the learning condition:
learned if $\max F(\text{offdiag}) < F(\text{diag})$

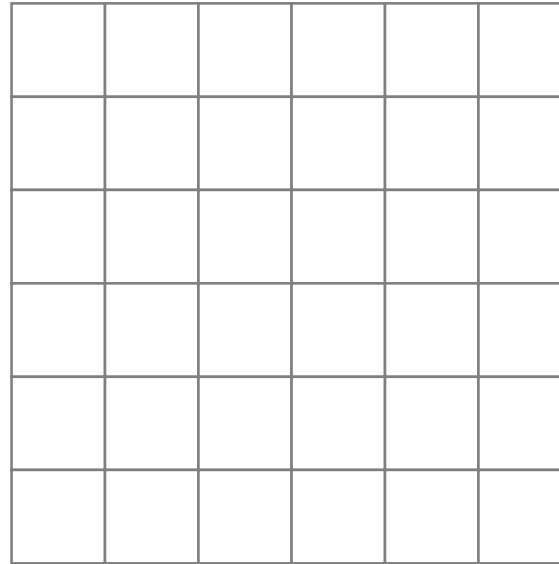
Schematic for Zipf: $\# \text{words} = \# \text{objects}$



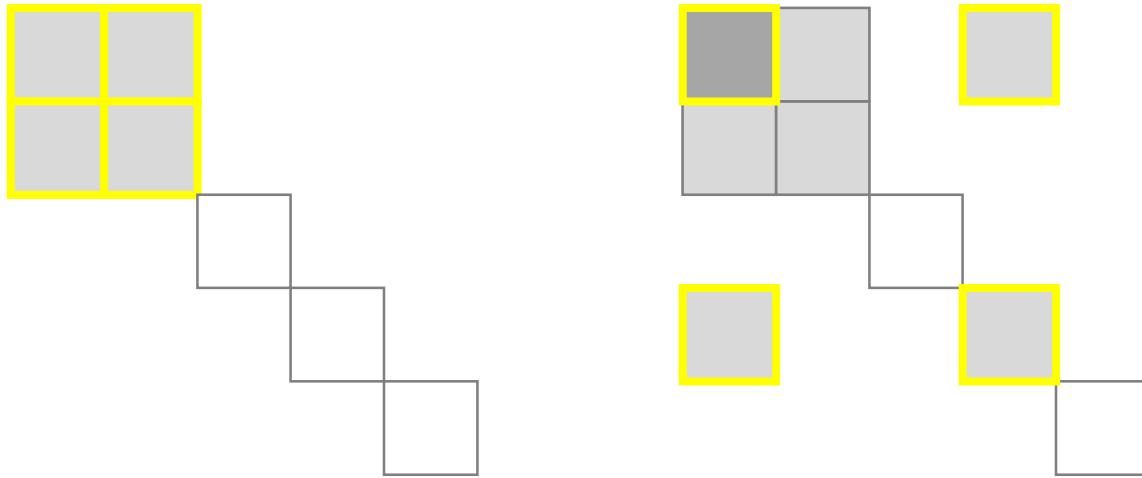
No words/objects alone

Best to the learning condition:
learned if $\max F(\text{offdiag}) < F(\text{diag})$

Memo



Sparse: $n = m \rightarrow$ no mislearning (easy?)

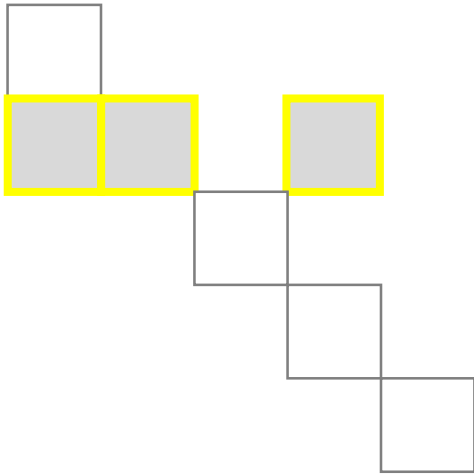


To be learned, at least 2 times
without overlap

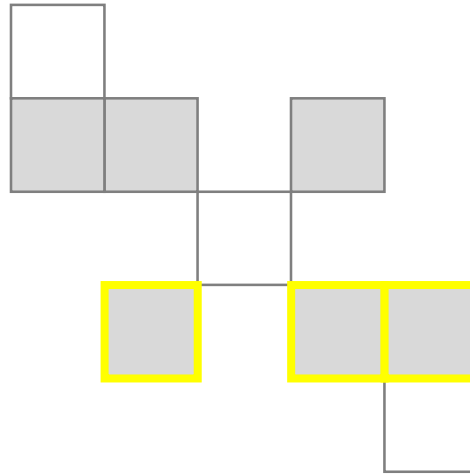
If more words/objects, first obs is easy;
many objects at the same time;
But it gives more overlap,
so more than 2 times is required

1 items learned

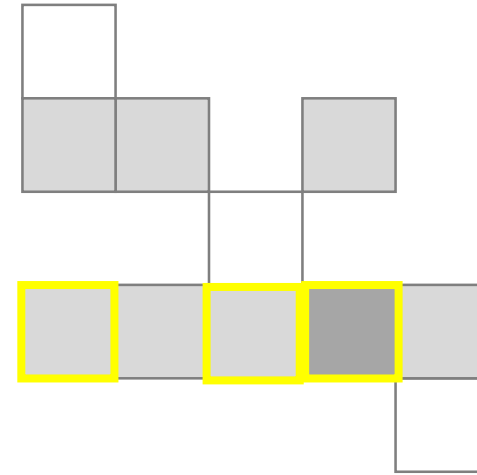
Sparse: $n < m \rightarrow$ learning hard?



0 items learned

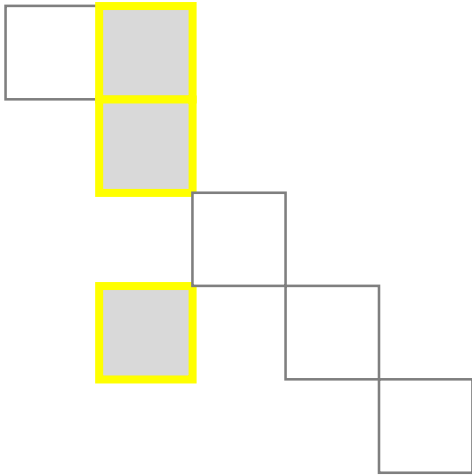


0 items learned

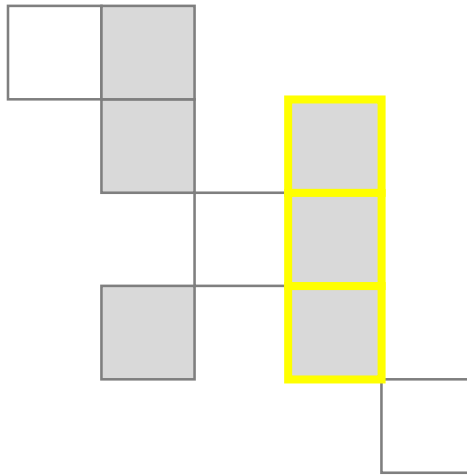


1 items learned

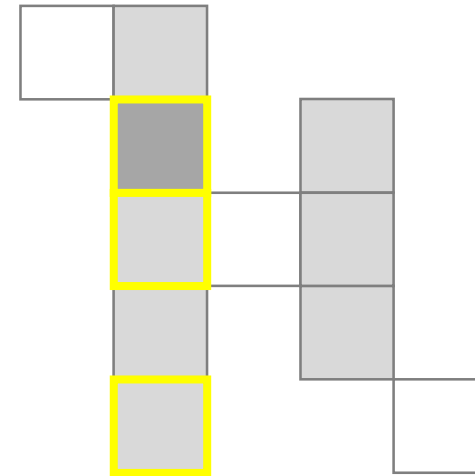
Sparse: $n > m \rightarrow$ learning hard?



1 items learned



0 items learned



1 items learned

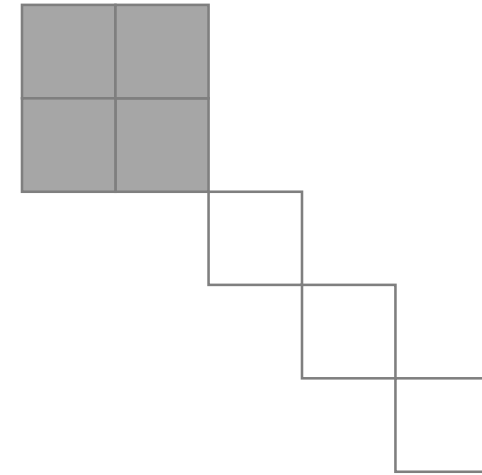
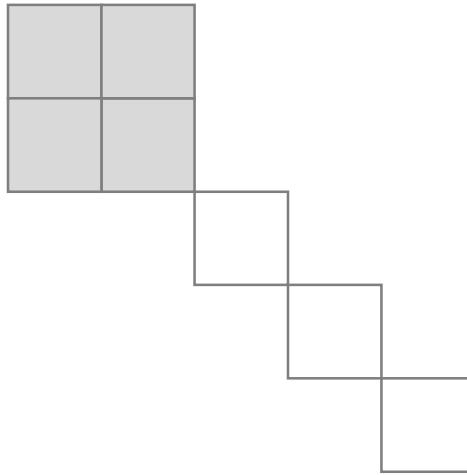
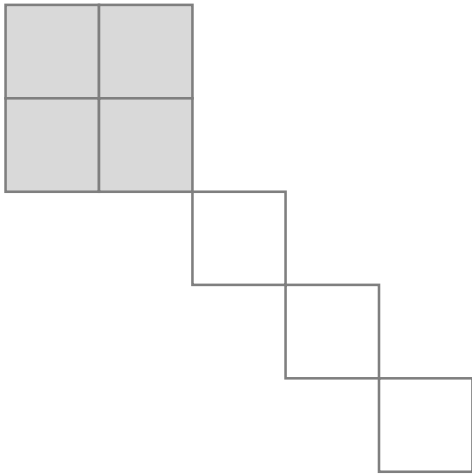
If uniform choice $\rightarrow$ Observing rare objects (easy)

2 times

If correlated choice $\rightarrow$ Observing same objects (hard)

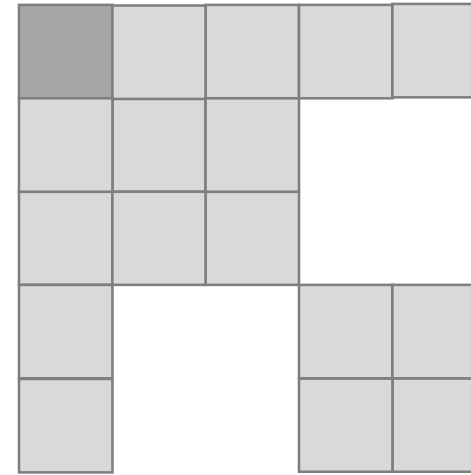
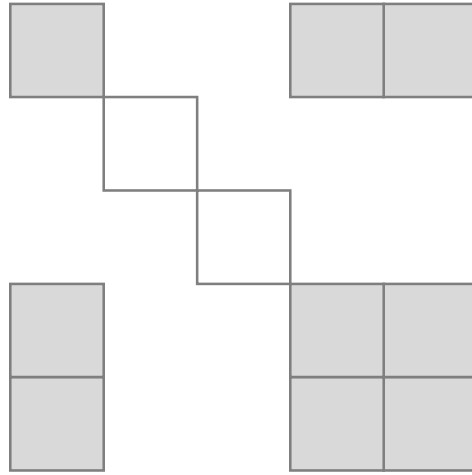
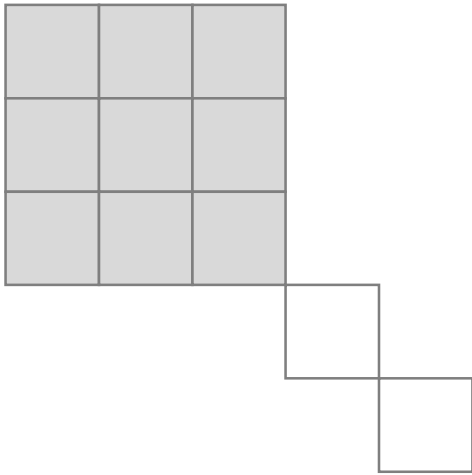
more than 2 times

Sparse, correlated



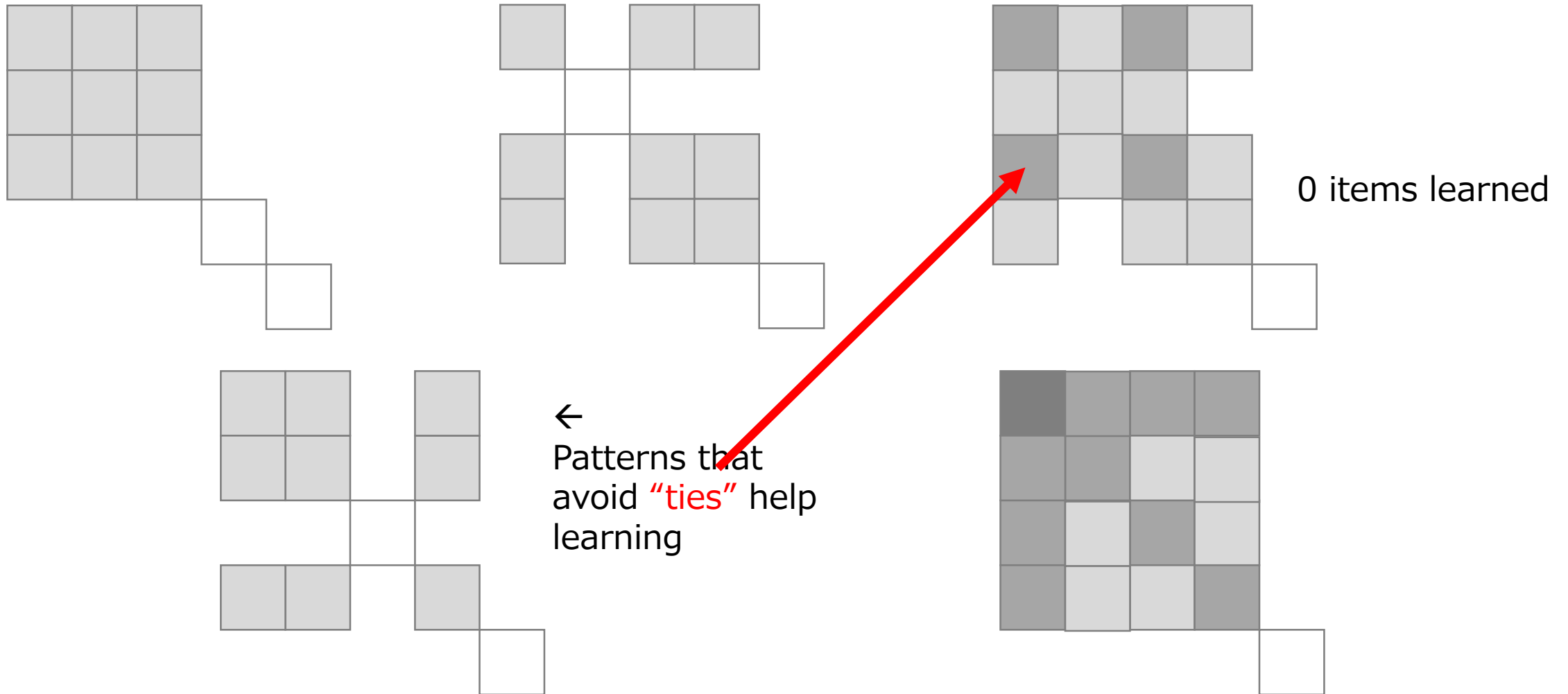
0 items learned

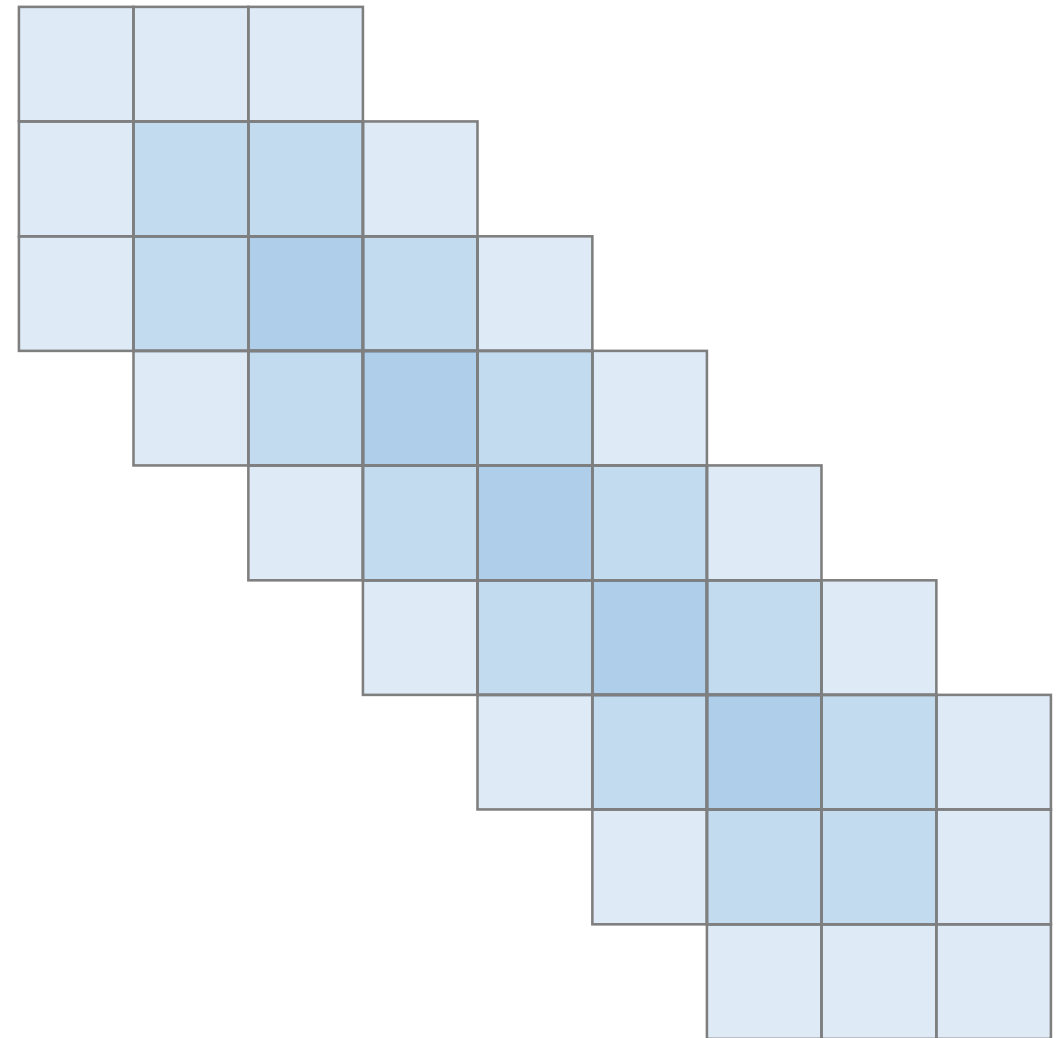
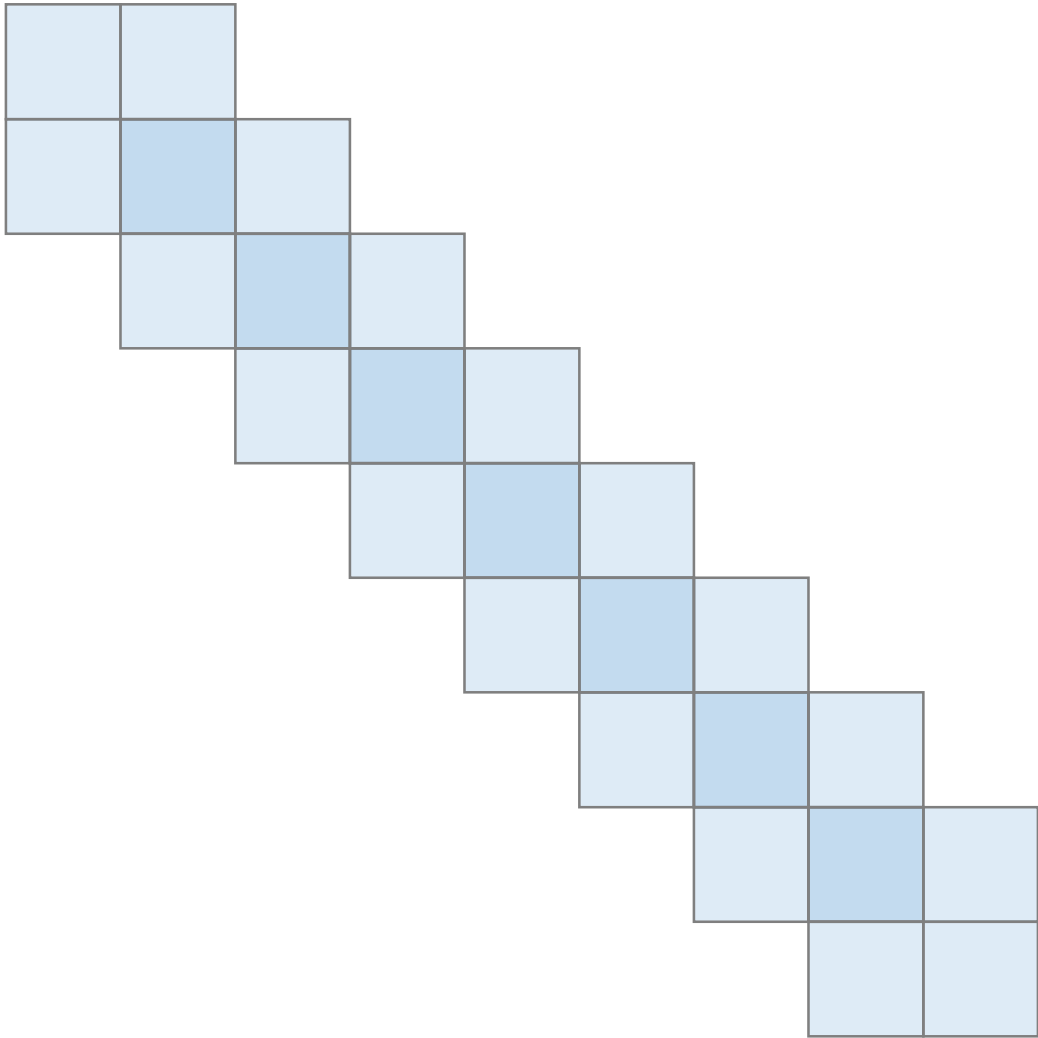
Dense



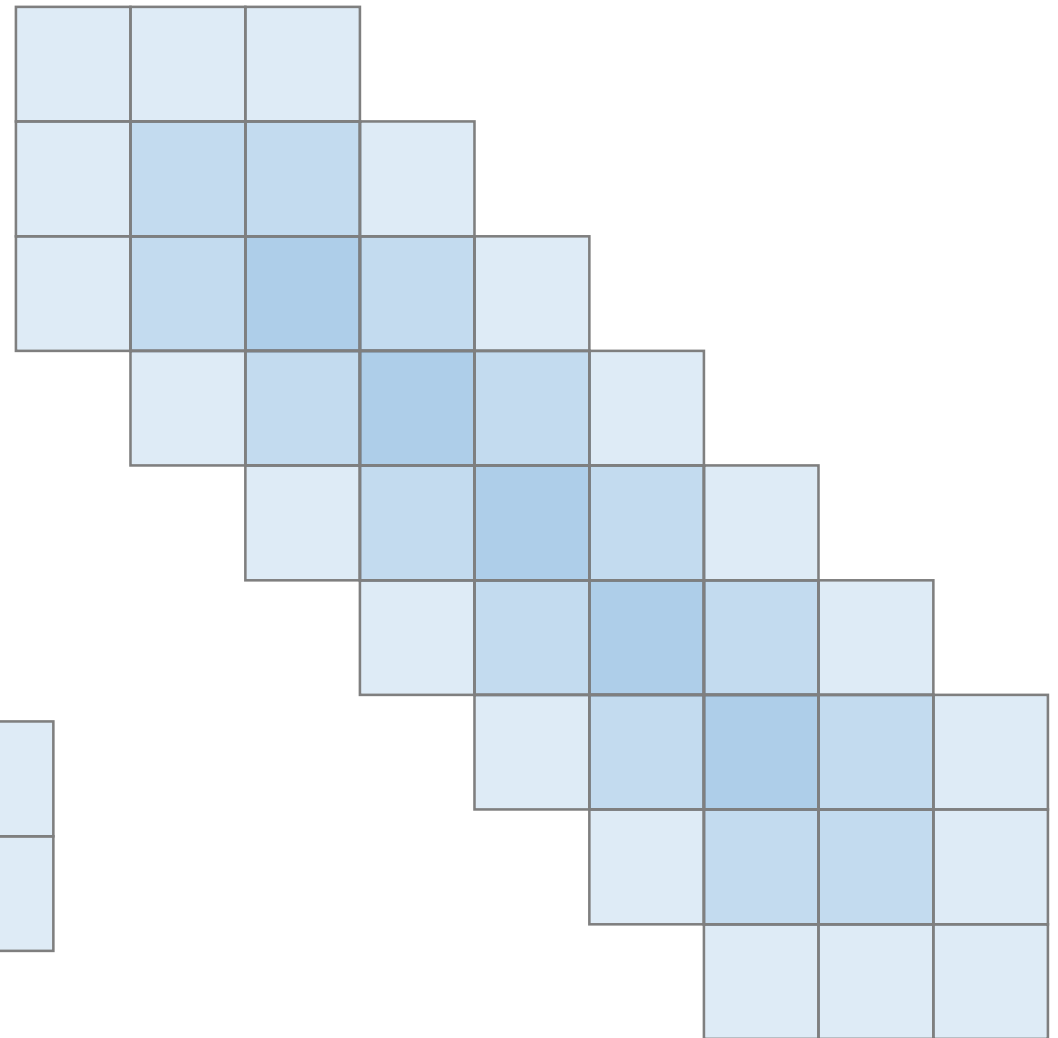
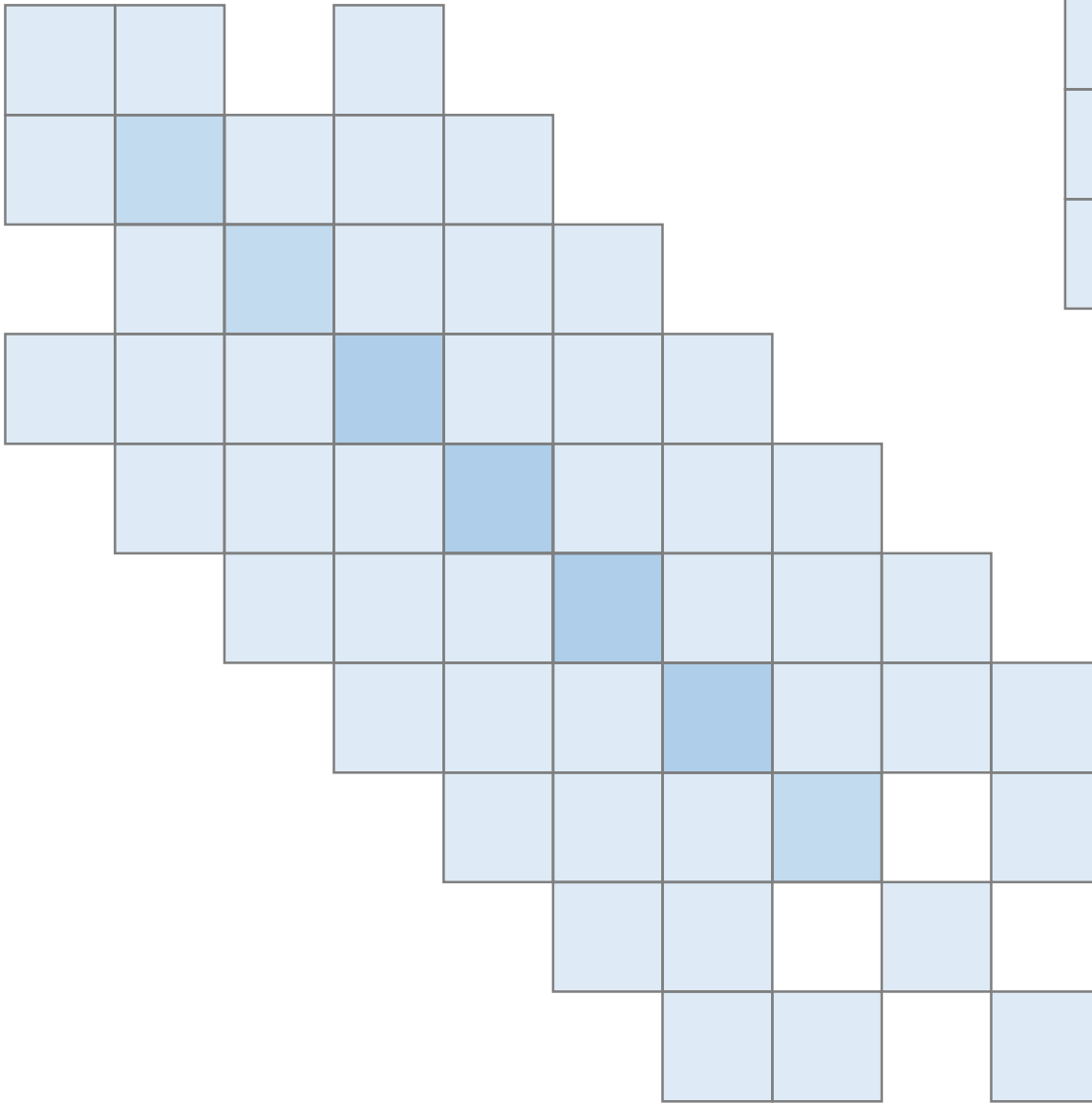
1 items learned

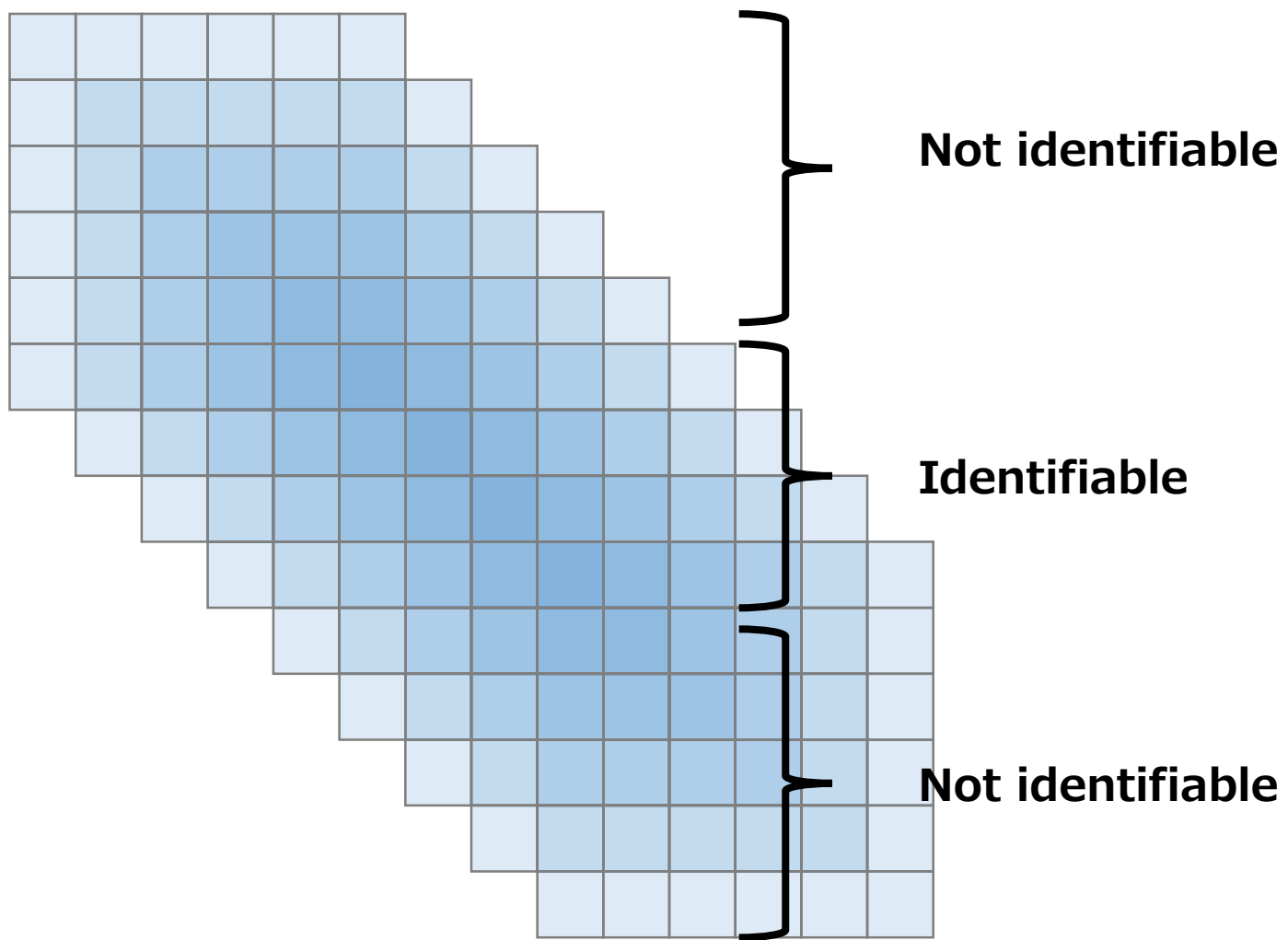
Dense, correlated





$n = m$, probably most efficient scheme
is like this



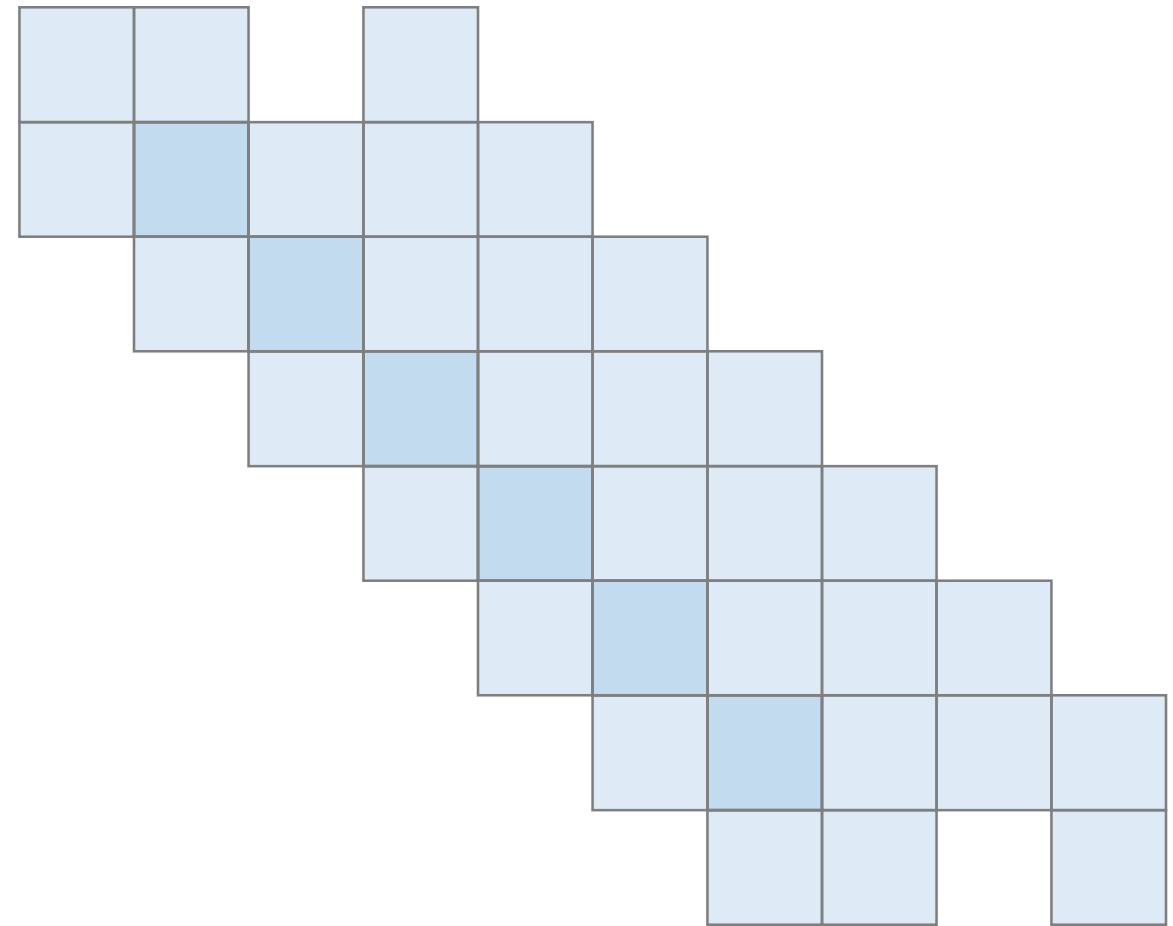
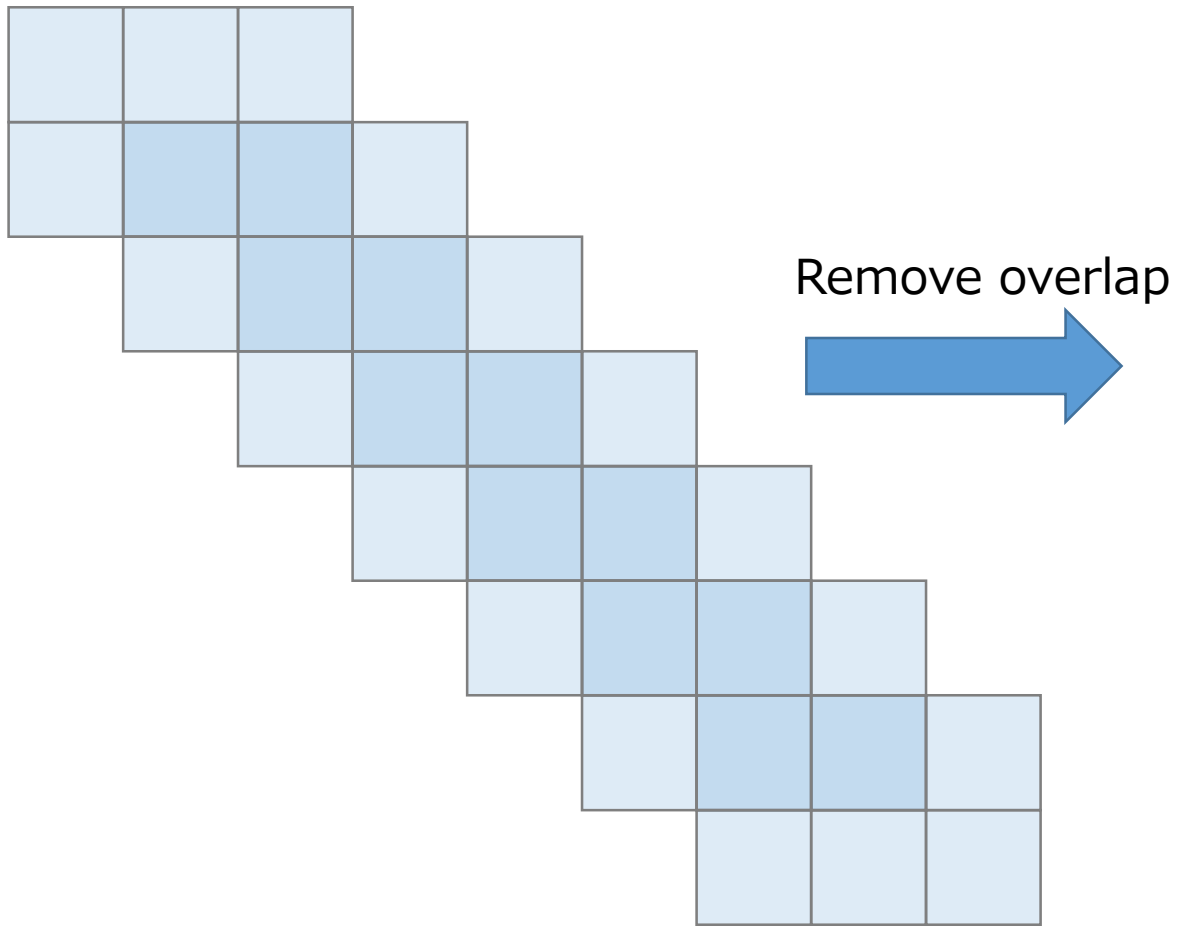


Overlaps removed

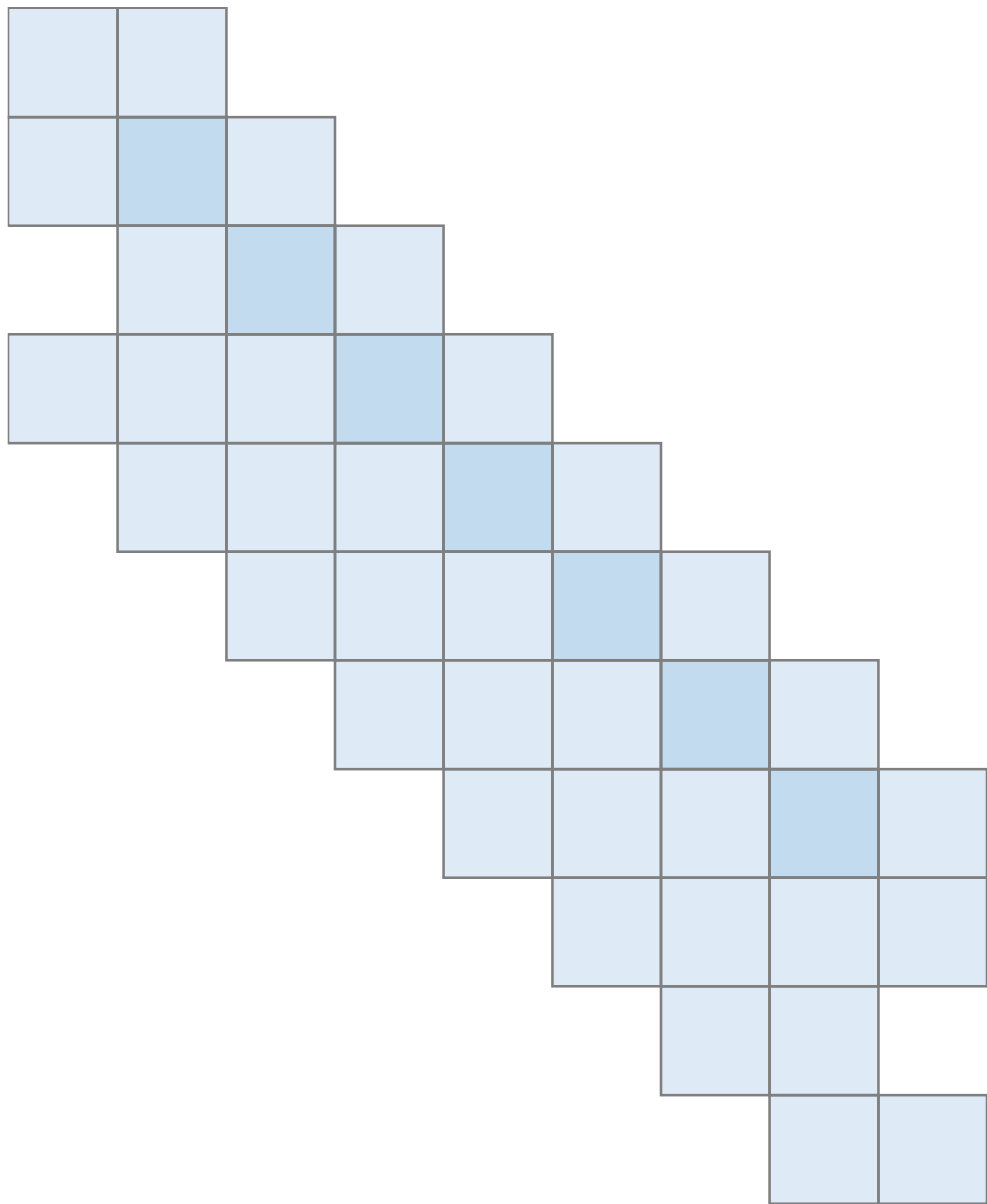
Not identifiable

Identifiable

Not identifiable

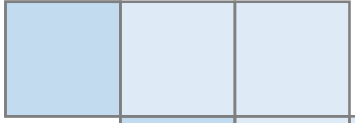


if $n \neq m$, the previous scheme is not available
But shifting the last to the next gives a solution
that follows the "same" scheme
Key idea behind is





$n = 1$



Each requires at least 2 observations

